SPEAR3 LLRF Upgrade

Subsystem Integration and Migration Plan

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This plan sets the order in which the subsystems of the SPEAR3 LLRF Upgrade are integrated into the running legacy RF station, I1 to I8, and works out for each integration the interfaces it must keep, its worklist of hardware, software and tests, and how it is reversed. It is based on the project status and decisions of 31 October to 8 November 2026, the upgrade design (Doc 0), the legacy documentation, the original drawings and design notes, the legacy EPICS source code, the decoded RF MPS PLC-5 program and the LLRF9 vendor software. Statements that still need confirmation are tagged **[O*n*]** and collected in the Open Items register.

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Revision History

Revision	Date	Scope of Change	Summary
R0	2026-11-11	Draft for review	Initial issue: layout and couplings of the legacy station, integration sequence I1–I8, the requirements of each integration (interfaces, entry criteria and fallback) with a worklist of hardware, software and tests, software plan, deferral and contingency, based on the project status on 8 November 2026, the decoded PLC-5 program and the LLRF9 software of April 2026. On 10 November the Interface Chassis lost the key switches and the beam-abort output, which nothing at SPEAR3 uses (Doc C). On 11 November the Fast Interlock Chassis's spare-fault input was traced to VXI Slot 5, the AIM's External MPS channel; from I2 the Interface Chassis takes the SPEAR MPS and orbit permits in parallel with Slot 5, which stays until I8, and drives that channel.

Open Items — Still To Be Confirmed

The statements below come from drawings, design notes and source code and have not yet been confirmed on the installed hardware or with the responsible engineer. Each is tagged **[O*n*]** where the plan relies on it; the page column links to that point.

ID	Item	What is needed	Needed before	Page
O1	Interface Chassis logic	The equations that form SCR ENABLE, KLYSTRON CROWBAR and the LLRF9 enable are written from our understanding of the system's required behavior [R45], not recovered from the panel's GAL22V10, whose equations are not documented. Review them with J. Sebek and the HVPS and SPEAR MPS owners, and compare them with the recorded behavior of the panel (I0-T1); each difference is kept or removed deliberately.	Interface Chassis logic design (I2)	p. 19
O2	Panel cross-connect lines	The PLC-5 program and the wiring diagrams give the direction, level (24 V) and use of every J2 and J3 line (Table 29). The drawings disagree on the terminal numbers at X530 and the panel schematic and the cross-connect diagram on the J3 pin order; check every line by continuity before the Interface Chassis cables are made.	Interface Chassis interface to the PLC-5 (I2)	p. 18
O3	Fast Interlock Chassis functions	Confirm on the hardware how the AIM's 23 outputs on J15 are used besides the six commands (by the drawings, the AIM's supplies to the chassis and twelve arc trip tests); whether any other AIM input depends on the chassis; which tunnel fibers feed the enabled AIM arc channels; and that the klystron-arc and transformer-arc triggers reach only the HVPS controller, as the drawings show.	Moving the arc fibers (I1); Interface Chassis interface to the AIM (I2)	p. 17

ID	Item	What is needed	Needed before	Page
O4	Legacy heartbeat	The program sets the heartbeat: a pulse one scan long every 50 ms plus a scan or two, stopped by a read error from any analog or thermocouple module until reset. Measure its period and duty cycle at the panel output and at the AIM input, and size the Interface Chassis watchdog from them.	Interface Chassis production design (I2)	p. 18
O5	Legacy IOC changes	Confirm that the legacy database can still be built and reloaded (last changed in 2021), whether the SNL programs can still be rebuilt (EPICS 3.14.11 for vxWorks 5.4, built on a Solaris host according to the release notes), and whether a test IOC can be set up.	Any legacy IOC change; the minimal coordinator (I6)	p. 12
O6	Slot 5 permit loop	Trace the SPEAR MPS and orbit-interlock wiring into VXI Slot 5 and check the source current budget for a second input in the Interface Chassis.	Interface Chassis installation (I2)	p. 18
O7	2021 LLRF9 setup	The vendor software of April 2026 holds the SPEAR3 configuration built since 2021—channel maps, tuner loops, a tuner interface to the legacy steppers, an HVPS loop and a station state machine—linked to PVs with the prefix SRFTEST. Recover what SRFTEST was, how the drive was switched, how the interlocks were wired and how turn-on was sequenced in the 2021 tests, and document the 2U “LLRF Window Interlock Status” chassis in rack B132-14, built in 2021, which turns a legacy permit into an enable for the LLRF9.	LLRF9 as drive source (I6)	p. 28
O8	RF MPS program	The decoded PLC-5 program of 2011 and 2012 is the reference. Upload the program in service, which carries the 2021 filament changes, and compare it with that reference; then confirm, rung by rung, that the ControlLogix program reproduces it, including the hardwired outputs, the global interlock sum and the heartbeat.	RF MPS (I4)	p. 24
O9	HVPS unit transfer	Write the list of connections transferred when the active HVPS unit is switched between a legacy and a new Hoffman box.	HVPS and PPS (I7)	p. 29

1 Purpose, Scope and Basis

The SPEAR3 LLRF Upgrade replaces the control electronics of the SPEAR3 RF station and keeps its high-power plant [R1]. Nearly all new hardware is built, the EPICS-side software has not been written, and the station must keep delivering beam between scheduled downs. This plan sets the order in which the upgrade subsystems of Doc 0 are integrated into the running station and works out, for each integration, the interfaces of the legacy station it must keep, the hardware and software it needs, its tests and its fallback. The high-power plant that is kept (klystron, HVPS power section, waveguide, cavities) and the proposed solid-state amplifier project [R22] are out of scope.

The plan is based on:

- the upgrade design, Doc 0 R4 [R1], which includes the transition interfaces of the Interface Chassis, the legacy functions of Table 18 and the integration sequence of this plan;
- the design requirements of the Interface Chassis and the Waveform Buffer, Doc C [R45];
- the legacy system: Doc L [R2]; J. Sebek's design notes and test report [R3]–[R6], [R9], [R23], [R24], [R34], [R37]; the original drawings and spreadsheets [R7], [R8], [R21], [R25]–[R30], [R41]; the legacy EPICS source code [R10]–[R19], [R32], [R36], [R42]–[R44]; the RF MPS PLC-5 program [R38]; the LLRF9 vendor software and documentation [R20], [R35], [R39], [R40];
- the project current status on 8 November 2026.

"No software has been written" refers to the EPICS side. The PLC ladder programs are taken as recorded: the RF MPS program written and tested on SPEAR3 without RF power [R1], and the HVPS program is still being completed. The PLC-5 program that the RF MPS program must reproduce is available in its 2011 and 2012 versions, decoded into readable listings [R38].

Terms.

Legacy station The RF station as it operates today: VXI LLRF crate, PLC-5 RF MPS, HVPS controllers, 1746-HSTP1 tuner controllers, Fast Interlock Chassis 340-308, AIM, RF Local Display Panel 340-310 and the variac heater supply.

RF Local Display Panel The legacy panel 340-310 in rack B132-12, also called the Local Control Chassis; its schematic SD-340-311-01 is titled "PEP-II RF System Local Panel" [R26]. Its logic forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort (§4.4).

Hoffman box The enclosure in B118 that is one HVPS controller: SLC-500, firing and regulator boards, trigger interconnect boards, power supplies, local control panel and PPS wiring [R1].

Integration One upgrade subsystem, or a group that goes in together, put into the operating configuration during a scheduled down; numbered I1 to I8.

Logical rack The address of a PLC on the Remote I/O link (1 to 3), not a physical rack (§4.1).

Monitor-only Installed and observed, with no output in any control or trip path.

Fallback The tested procedure that restores the previous configuration.

2 Summary

1. **Eight integrations in a set order** (Table 4): I1 the ARC4 and the two LLRF9 units, monitor-only; I2 the Interface Chassis and Waveform Buffer in place of the Fast Interlock Chassis and the RF Local Display Panel; I3 tuner control by LLRF9 Unit 1 and the Galil, while the RFP still drives; I4 the RF MPS; I5 the heater supply under RF MPS control; I6 the LLRF9 as drive source, in parallel with I4 and I5; I7 a new HVPS controller, a complete Hoffman box, with the PPS Interface Box, unit by unit; I8 the retirement of the legacy station, when B132-IOCRF moves to new hardware.
2. **Monitor-only work first.** The ARC4 and the LLRF9 measurement go in while the Interface Chassis is in production; neither touches the protection chain, and the ARC4 can trip only through the Interface Chassis, from I2 on.
3. **I2 is the hardest integration.** While the AIM, the PLC-5 and the HVPS controllers are still in service, the Interface Chassis must present what the two legacy units present today—the AIM's J15 cable, the panel's cross-connect lines to the PLC-5, the three HVPS fibers and the buffered HVPS voltage and current—and form the panel's outputs with logic written from the required behavior of the system, because the panel's own equations are not documented (§6.3). These transition interfaces are not in the Interface Chassis specification [R3].
4. **The LLRF9 goes in in two steps.** I3 moves tuner control to the LLRF9 and the Galil without a change to the legacy SNL programs. I6, the LLRF9 as drive source, needs the Interface Chassis and the minimal

coordinator in B132-IOCRF, which runs on the VXI CPU until I8; it goes in when that software is ready, so the legacy build environment (O5) gates only I6. The vendor software already holds a station state machine and an HVPS loop on the drive power [R39], which the coordinator configures and commands rather than writes.

5. **I4 before I5** lets the heater go straight to its final control by the RF MPS. I7 comes last before retirement, one HVPS unit at a time; I8 needs all the others, a separate reference distribution, the full coordinator and a home for every legacy function (§7).
6. **Software is the critical path.** Keeping the legacy PV names and SNL programs limits every integration except I6 and I8 to its subsystem IOC and small database changes (§8). The running station is the test bed: one integration per down, monitor-only first, and a tested fallback (§5.2).
7. **A worklist per integration.** Each integration carries a numbered list of its hardware, software and test items with their dependencies (§6); the entry criteria refer to these items.

3 Project Status

Table 1: Status of the upgrade subsystems, 2 November 2026.

Subsystem	Hardware	EPICS-side software
LLRF9 (two units)	Installed in B132; RF cabling to be completed in the coming weeks. One unit will run the four tuner loops and drive the Galil directly.	Vendor IOC, upgraded to EPICS 7 in June 2026 [R1]; the vendor software of 29 April 2026 holds the SPEAR3 channel maps, tuner loops, HVPS loop and station state machine [R39]
Galil DMC-4143	Chassis built; tested on the SPEAR3 tuner motors under an EPICS control written by M. Dunning [R1], [R33]. That control is not carried forward, because the LLRF9 drives the Galil directly.	None needed beyond the LLRF9
ARC4	Assembled; parts from the same vendor as the existing arc-detection hardware; accepts the existing tunnel fibers; ready to install	None; its signals are read through the Interface Chassis
Interface Chassis and Waveform Buffer	One chassis with two boards; prototype ready for review, then production	Waveform Buffer tested standalone in April 2026 [R1]; none for the Interface Chassis
RF MPS (ControlLogix)	Assembled; online for software development; EPICS communication working; ladder program written, to be compared with the decoded PLC-5 program (O8)	None
HVPS controller (new Hoffman box)	CompactLogix online for software development, EPICS communication working; FCOG1200 firing boards ordered; regulator board being redesigned	None
Heater supply (Genesys AC)	Running on a test stand with a load; external 60 Hz sync circuit in design	None
PPS Interface Box	Design of the initial version done; documentation mostly done; ADSO approval and the final design still needed	Not applicable
Control software	TID software review completed	No EPICS-side software written

The critical path has moved from hardware to software, reviews and approvals, and installation windows. The firing boards, the regulator and the heater sync circuit are as recorded in Doc 0 [R1].

4 The Legacy Station

4.1 Physical Layout

Table 2 gives the location of the legacy equipment that the integrations touch [R6].

Table 2: Racks of the legacy station.

Rack	Equipment
B132-11	VXI crate, whose slot-0 CPU runs the IOC B132-IOCRF; Fast Interlock Chassis 340-308 [R2]; drive amplifier; RF patch panel [R8]
B132-12	RF Local Display Panel 340-310; above it a spare Fast Interlock Chassis 340-307, a slightly different version, not in use
B132-13	RF MPS PLC-5 (main crate and a second crate below it); tuner stepper chassis 340-315; the two LLRF9 units
B132-14	Cross-connect blocks, among them X530 for the panel's connectors J2 and J3 and X15 for the waveguide pressure switches; a spectrum analyzer and, above it, the LLRF Window Interlock Status chassis of 2021 (O7)
B132-15	Klystron filament heater chassis (variac) and the main focus and bucking coil supplies
B118	HVPS controllers, one Hoffman box per HVPS unit

4.2 Couplings

The legacy station is held together by four couplings (Figure 1):

1. one Allen-Bradley Remote I/O (RIO) daisy chain, through which every PLC datum reaches EPICS (§4.3);
2. one hardwired protection chain—AIM, Fast Interlock Chassis, RF Local Display Panel and the HVPS fibers—and the permit inputs of VXI Slot 5 (§4.4);
3. one RF drive source, the RFP, with the three IQA modules as the measurement front end (§4.5);
4. one supervisory IOC, B132-IOCRF, whose SNL programs and alarm tree coordinate the station (§4.6).

The heater and the focus and bucking coil supplies hang on the PLC-5, which switches them and reads and trips on their voltages and currents [R7], [R13], [R21]. An integration can be done inside the running station only where its interfaces to these couplings stay the same, or where the new subsystem reproduces them.

4.3 The Remote I/O Chain

The VXI slot-1 Allen-Bradley 6008-SV1R scanner, mounted through an HP E1407A adapter, is the master of a single RIO link [R2]. Its configuration addresses the three PLCs as logical racks [R11], [R32]; this plan names them after their devices (Table 3).

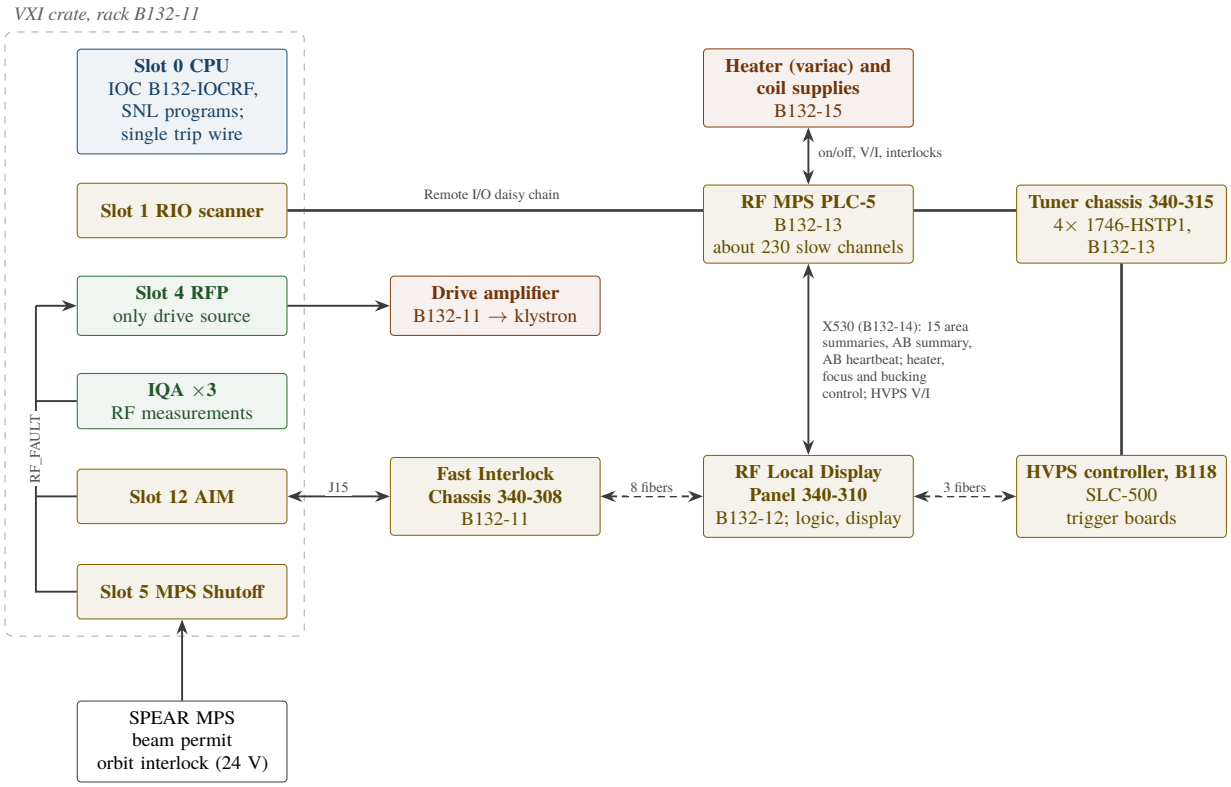


Figure 1: Couplings of the legacy station, with the racks of Table 2. The Remote I/O link is one daisy chain from the VXI scanner through the RF MPS PLC-5 and the tuner chassis to the HVPS controller. Every fast RF-off path ends on the RFP through the open-collector backplane line RF_FAULT. The AIM, the Fast Interlock Chassis, the RF Local Display Panel and the HVPS controller form one hardwired protection chain; the panel's logic forms SCR ENABLE, KLYSTRON CROWBAR and a beam abort on BNC J6, which is not cabled. The PLC-5 controls the filament heater and the focus and bucking coil supplies; the coil-supply faults also enter the panel's logic. Dashed arrows are fibers.

Table 3: Nodes of the legacy RIO link.

Node	Logical rack	Device and adapter	Physical rack
PLC-5 node	3	RF MPS PLC-5 through a 1771-DCM; first node	B132-13
Tuner node	2	SLC-500 tuner chassis 340-315 with four 1746-HSTP1 modules; middle node	B132-13
HVPS node	1	HVPS SLC-500 through a 1747-DCM; last node	B118

The chain runs from the scanner to the DCM in the main PLC-5 crate, on to the tuner chassis and, through a terminal box above the HVPS termination tank, to the HVPS SLC-500 in B118; the second PLC-5 crate hangs on the main crate's own link [R6]. Consequences:

- Every PLC datum and command reaches EPICS only through this chain and B132-IOCRF.
- A powered-down node loses only its own data, but disconnecting a node's terminals cuts every node downstream: replacing the PLC-5 or the tuner chassis means splicing the chain through, and removing the HVPS node means moving the terminating resistor.

- The PLC-5 node and the HVPS node behave differently when they stop answering. The DCM record that reads the PLC-5 never reprocesses its records on a link change, because its LOCA field is not set: while the IOC runs, the PLC-5 values freeze, the adapter and DCM records go to MAJOR, the station status shows HWERROR and nothing trips; after an IOC reboot without the node the PLC-5 records stay undefined and the state machine refuses to turn the station on. The scanner driver instead sends the HVPS records to INVALID, which turns the station off [R42]. The PLC-5 itself does not depend on the link (§4.8), so its node can be taken off with the IOC running, as the I4 entry test does (§6.5); the node's records are then re-pointed as in §8.3.

4.4 The Hardwired Protection Chain

The chain is documented at the level of connectors and fibers in J. Sebek's notes [R6], [R34] and in the original drawings [R26]–[R30]:

- **Fast Interlock Chassis 340-308.** Connector J15 is cabled to J3 of the AIM (VXI slot 12). Of its 25 lines, 23 are driven by the AIM. Six of these are the commands FILAMENT_ON (filament and focus request), SOLENOID_ON (station online request), HVPS_ON, BEAM_ABORT, FILAMENT_TIMEOUT (filament timeout bypass) and FAULT_RESET, which the chassis converts to fibers to the panel. The two inputs, the AB summary fault (pin 10) and the AB heartbeat (pin 23), arrive as fibers from the panel and drive Schmitt-trigger inputs on the AIM [R6], [R28]. The chassis also holds a detector for the klystron forward power, with RF input J32 and output J31 on the rear panel, cabled to the PLC-5, and a front monitor output J16 that is not cabled; an arc-detection front end that is largely disconnected, whose 37-pin connector J2 is cabled to J4 of the AIM; and a spare-fault input, cabled to J3 of the VXI Slot 5 MPS Shutoff module, that reaches the AIM through the same cable as its External MPS channel [R6], [R27].
- **RF Local Display Panel 340-310.** Eleven fibers (Appendix C): the six commands and the AB summary and heartbeat to and from the Fast Interlock Chassis [R29], and three to and from the HVPS controller. Two 25-pin connectors, J2 and J3, are wired to cross-connect block X530 in rack B132-14 [R6], [R7], [R30]. They carry the AIM's requests to the PLC-5 as 24 V lines, the PLC-5's 15 area summaries, its global interlock sum (the AB summary) and its heartbeat, the filament, focus and bucking-coil supply control and interlocks, and the HVPS voltage and current, which the panel buffers for the PLC-5's collector-power calculation (Table 29). The panel displays the station status and carries the station online/offline and local/remote key switches [R26].
- **The panel's logic.** A programmable logic device (GAL22V10) forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort from the HVPS on request, the VXI and HVPS beam aborts, the station online key switch and four lines from the PLC-5, whose logic the program gives [R38]: the main-focus and bucking-coil summaries, the waveguide air-pressure summary and the collector-power summary. A coil summary is on while the filament is requested and the supply's voltage and current are inside their limits; the main-focus summary also drops when the supply's internal contact (PLC-5 input 025) opens, while the bucking-coil contact (input 026) is not used. Two flip-flops latch the beam abort and the crowbar until reset [R26], [R34]. The beam abort leaves on a rear-panel BNC connector (J6), which is not cabled at SPEAR3 [R26], [R46]. The equations of the logic are not documented, so the Interface Chassis does not copy them (O1).
- **HVPS fibers.** Three fibers run between the panel and the trigger interconnect boards of the active HVPS controller in B118, through a switchbox that selects the unit [R6], [R29], [R31], [R34]. SCR ENABLE (lit to enable) goes to the Right Side Trigger Interconnect board (SD-730-793-07), which passes it to the Left Side board; while it is dark both boards inhibit the triggers of phases A and C, and B^+ and B^- keep firing so that the filter inductors discharge [R4], [R5]. KLYSTRON CROWBAR (lit for no crowbar) goes to the Left Side board (SD-730-793-08); when it goes dark the crowbar fires and the triggers are

inhibited. STATUS comes from the Left Side board and is lit while the controller's "24 V fault/enable" output is high and no crowbar is commanded; the panel receives it as the "HVPS beam abort" [R4], [R26].

- **Arc triggers in the HVPS controller.** A Pearson current transformer in the termination tank feeds the klystron-arc trigger input of the Left Side board, and the transformer-arc trigger enters the Right Side board. Either one fires the crowbar and inhibits the triggers in hardware; the SLC-500 only latches both as status bits [R4], [R5], [R14], [R25]. Neither passes through the Fast Interlock Chassis or the panel, which sees them only as a dark STATUS fiber.
- **VXI Slot 5 ("MPS Shutoff").** The SPEAR MPS beam permit and the orbit interlock (24 V each) [R3]; dropping either asserts RF_FAULT [R2]. Its J3 output drives the spare-fault input of the Fast Interlock Chassis, which the AIM holds as its External MPS channel. The module and its wiring are traced under O6.
- **RF_FAULT.** An open-collector VXI backplane line. The AIM, the IQAs, Slot 5, the CLK module and the RFP can assert it; the RFP then removes the drive [R2].

Beam abort. The state machine forces a beam abort through the AIM whenever it turns the station off, enters TUNE or switches the direct loop, and resets it when the station reaches ON_CW with no faults [R17]. The panel's logic combines it with the HVPS STATUS and its other inputs, by equations that are not documented (O1), and drives J6. The output was made for the PEP-II beam-abort system [R46]; at SPEAR3 J6 is not cabled, and the SPEAR MPS takes no status from the RF in hardware: it dumps the beam by removing the RF permit through Slot 5 [R3]. The beam abort therefore stays inside the station: the AIM's latch, which the state machine reads, and the panel's lamps.

> Every fast RF-off path of the legacy station ends on the RFP. High voltage is removed in hardware through the panel's SCR ENABLE and KLYSTRON CROWBAR fibers or inside the HVPS controller. A SPEAR MPS or orbit trip reaches the HVPS only through the RF cut and the state machine, unless the PLC-5's collector-power protection acts first: after any trip the state machine spends about 1 s in TUNE and 5 s ramping the HVPS down before it removes the triggers and the RF switch [R17].

4.5 The RF Drive and Measurement Path

- The RFP is the only drive source. The 2021 cable record lists the RFP output and the drive-amplifier input as separate, marked patch cables next to the LLRF9 reference, which came from output RF3 of the VXI CLK module [R8]: the drive source is selected by a patch cable.
- The three IQA modules measure the 24 RF signals. The legacy loops and alarms use their data: drive power for the HVPS loop, probe and forward phases for the tuner loops, and reflected and forward powers for the alarms.
- In 2021, 20 dB couplers with about 0.4 dB insertion loss were inserted upstream of the legacy system for the nine signals the LLRF9 monitored—klystron forward, cavity forward and cavity probes—and the legacy calibration constants were changed to compensate [R9].

4.6 The Supervisory IOC B132-IOCRF

- The VXI slot-0 processor runs the IOC B132-IOCRF under vxWorks. It loads one expanded database, `srf1.db`, restores saved settings, configures the RIO scanner from `config.ab` and starts nine SNL program instances: the state machine, four tuner loops, the HVPS loop, the DAC loop, a calibration sequence and the message logger [R10]. It stays the station IOC and the coordinator, and moves to new hardware at I8.

- The state machine trips on one summary, `SRF1:STNOFF:SUMY:STAT`. It collects the VXI latch, the HVPS off summary, the PLC-5's online bit (the panel's online key and the station-online command), the AB summary latch and the PLC-5's own station summary bit; turn-on is blocked by a similar summary, `SRF1:STNON:SUMY:STAT` [R17], [R19]. Cavity, tuner and temperature alarms reach only the station status display.
- A database change is a text edit and a reboot. An SNL change needs the EPICS 3.14.11 vxWorks cross-development environment, whose availability is not confirmed. **[O5]** The release was built on Solaris for vxWorks 5.4, in AFS and later in NFS, and its last changes, in 2021, were to the database only [R43].
- The SNL bindings that matter for the integrations, among them the HVPS loop's dependence on the RFP [R16], are listed in Appendix B.

4.7 Data the PLC-5 Provides to Other Subsystems

The PLC-5 is the station's slow data system as well as an equipment-protection PLC. About 230 template instances in the legacy database take their data from it (Appendix B): about 110 temperatures, about 40 interlock bits, about 30 analog channels and about 50 summary, module and digital status points [R13], [R14]. Several belong to other subsystems: the tuner potentiometer positions (`SRF1:CAVnTUNR:POSN`) and temperatures; the stepper drivers' bulk supply voltage (`SRF1:CAVTUNR:VOLT`), whose latch feeds the tuner summary [R13], [R19]; filament voltage and current with their PLC-5 trip limits, and the main focus and bucking-coil supply voltages and currents; klystron and circulator ion-pump currents and window temperatures; and the HVPS voltage and current as seen by the PLC (the collector-power inputs) and the HVPS oil temperature [R12]. Every integration that touches the PLC-5, the tuner, the heater or the HVPS checks these channels.

4.8 The RF MPS Program

Four RSLogix 5 projects of the PLC-5 program—an undated one with the address descriptions and edits saved in November 2011 and February 2012—have been decoded into readable listings [R38]. Their logic is the same apart from two limit changes. The 2021 filament changes are not in them, so the program in service has to be uploaded and compared (O8). The program shows:

- **EPICS link.** The PLC-5 writes eight 64-word pages to the DCM in turn; word 3 of each page holds the page number, which is the table number that the legacy records address. A page is resent after a transfer error, and the block read back from the DCM is used by no rung. Nothing in the program depends on the link, so the PLC-5 runs the same with or without the IOC.
- **Global interlock sum.** One bit is both the station summary that EPICS reads (`SRF1:STN:SUMY:PLC`) and the AB summary line to the AIM. It requires the four cavity summaries, the waveguide-load, waveguide air-pressure and circulator summaries, an input labeled "MPS" on the drawings, and—while the online bit (`SRF1:STN:LOCAL:ON`, the online/offline-mode and klystron-focusing lines both on) is set—the klystron group (klystron water, temperatures, vacuum and air flow; filament; focus; collector power) and the flow, vacuum and temperature summaries. Every fault latches until the panel's reset line, once its condition has cleared.
- **Heartbeat.** A 50 ms timer turns the heartbeat output on for one scan each time it expires. It stops when any of the 16 analog and thermocouple modules reports a read error, until reset; the DCM is not involved.
- **Collector power.** The beam power from the HVPS voltage and current buffered by the panel, minus the RF output power from the forward-power detector input "RF PWR LEVEL" [R41], trips above 1 200 kW for 0.7 s. HV detected and RF detected are thresholds on the same inputs, at about 9.7 kV and 9 kW.
- **Filament.** The warm-up sequence of §6.6, the trips on the heater voltage and current, and the window-air blower, which runs while the filament is on and for 600 s after.

- **Cavity vacuum.** The analog cavity-vacuum readings trip on eight pressure steps with delays from 900 s down to 1 s; these trips latch bits that only EPICS reads and are not in the global sum.

5 Integration Plan

5.1 Sequence

The project has set the order of the integrations (Table 4); Figure 2 shows which links in the order are technical and which are only planned. Monitor-only work comes first: the ARC4 and the LLRF9 measurement (I1) go in while the Interface Chassis is still in production, and neither touches the protection chain. The Interface Chassis (I2) follows; it gives the ARC4 its trip input and the LLRF9 its enable, so neither needs an interim path in the legacy chain. Tuner control (I3) moves to the LLRF9 and the Galil while the RFP still drives, without a change to the legacy SNL programs. The RF MPS (I4) goes before the heater (I5), so that the heater goes straight to its final control. The LLRF9 as drive source (I6) waits only for its software and runs in parallel with I4 and I5. The HVPS controller and the PPS box (I7) wait for the PPS approval, and the legacy station is retired (I8) when nothing depends on it.

Table 4: Integration sequence.

Step	Integration	Replaces	Hard prerequisites	Fallback
I0	Preparation (§6.1)	—	—	—
I1	ARC4 and LLRF9 measurement, monitor-only (§6.2)	Disconnected optical arc front end	Arc fibers checked (O3); LLRF9 hookup as a controlled change; SW8	None needed
I2	Interface Chassis and Waveform Buffer (§6.3)	Fast Interlock Chassis; RF Local Display Panel	Production unit with the transition interfaces; O1 to O4; SW7; baseline test of the legacy chain; failure-mode analysis	Reinstall the Fast Interlock Chassis and the panel
I3	Tuner control by LLRF9 Unit 1 and the Galil (§6.4)	1746-HSTP1 tuner control	I1; SW2; legacy tuner loops off with their alarms adjusted	Motor cables back to the SS2000 drivers
I4	RF MPS (§6.5)	PLC-5	Legacy-faithful program (O8); SW1 and SW5; legacy database change; RIO splice; trip test	PLC-5
I5	Klystron heater supply under RF MPS control (§6.6)	Variac stage	I4; 60 Hz sync circuit; SW3; acceptance on the test stand; a long down	Variac chassis
I6	LLRF9 as drive source (§6.7)	RFP as drive source	I2; SW9; O5 to O7; interlocked RF transfer switch	Transfer switch to the RFP
I7	New HVPS controller and PPS Interface Box (§6.8)	Hoffman box of the HVPS controller and its PPS wiring, unit by unit	ADSO approval; PPS certification; the new box; SW6; O9	Other HVPS unit
I8	Legacy retirement (§6.9)	VXI crate; RIO link; tuner chassis	I1 to I7, both HVPS units; SW10 on the new hardware of B132-IOCRF; reference distribution; a home for every legacy function	None practical

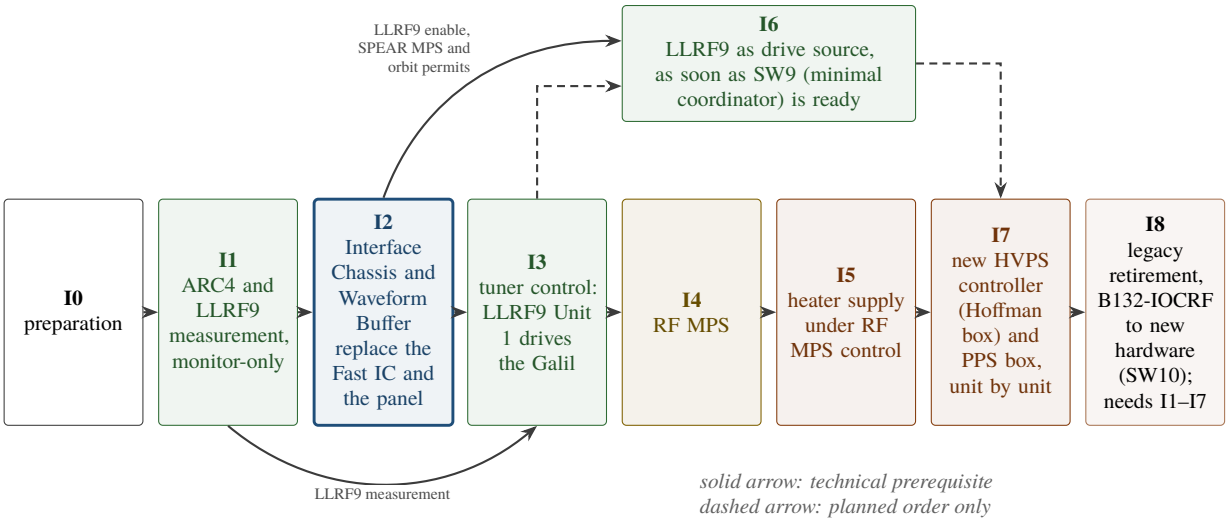


Figure 2: Integration sequence. Solid arrows are technical prerequisites, dashed arrows only the planned order. I3 needs the LLRF9 measurement of I1, and I5 the RF MPS of I4; I6, the LLRF9 as drive source, needs I2, waits only for its software and runs in parallel with I4 and I5; I8 needs all the others.

5.2 Rules for Every Integration

1. **Legacy names are the interface.** A replacement keeps the PV names, units, scaling and alarm limits of the records it replaces until the full coordinator takes over; the HVPS keeps its names for good.
2. **No new permits.** A new element in a permit path can remove a permit but never create one. Where it generates a permit itself, as the Interface Chassis does from I2 on, a failure-mode analysis and an equivalence test against the recorded legacy behavior come first.
3. **Monitor-only first** for a run period, where the subsystem allows it.
4. **One integration per window**, so that any new behavior has one cause; two integrations that need the same person are not scheduled in the same window.
5. **Tested fallback.** A written fallback procedure is tested before the window, and the legacy hardware stays intact until the next integration has run without it.
6. **Configuration control.** Every hybrid configuration is recorded: connections, calibration constants, database version, jumpers and bypasses.

6 Integration Requirements

Each integration is described under the same headings: scope; interfaces to the legacy station; a worklist of hardware, software and tests; entry criteria; fallback. The software work packages SW0 to SW10 are listed in Table 19.

6.1 I0: Preparation

Alongside the first integrations, without effect on beam (Table 5). The baselines of I0-T1 and I0-T3 are the references against which I2 and I4 are accepted. In every worklist, H marks hardware, S software and T tests and records; the last column names the open items, work packages and other items each one depends on.

Table 5: Worklist of I0.

Item	Work	Depends on
I0-H1	Production review of the Interface Chassis and the Waveform Buffer with the requirements of §6.3 and Doc C [R45] (A6)	O1 to O4
I0-S1	Decisions D1 to D11 recorded (A1); PV naming policy, legacy compatibility rules and one interface control document per integration (SW0)	—
I0-S2	Legacy IOC owner named; test IOC set up; procedure to edit, build and reload `srf1.db` written and rehearsed; SNL build checked (SW1, A2)	O5
I0-S3	The LLRF9 vendor software of April 2026 reviewed on a spare unit: its links to legacy PVs, which carry the prefix SRFTEST, its HVPS loop, tuner interfaces and state machine [R39] (A17)	O7
I0-T1	Baseline of the legacy protection chain: the panel's response to every input that can be set without high voltage, the three HVPS fibers, the HVPS controller's arc triggers as seen on STATUS, the heartbeat at the panel and at the AIM, and the request line at X530 that each AIM command drives (A3, A5)	O4
I0-T2	The PLC-5 program in service uploaded and compared with the decoded 2012 program [R38] (A16)	O8
I0-T3	The PLC-5 trip matrix run on the PLC-5 in service (A13)	I0-T2
I0-T4	Work that needs no window, started early: the Galil bench test (I3-T1), the heater test-stand acceptance (I5-T1), the PPS box approval steps (A14)	—

6.2 I1: ARC4 and LLRF9 Measurement, Monitor-Only

Scope. The ARC4 and the two LLRF9 units go in with no output in any trip or control path. The ARC4 chassis is assembled, ready to install and accepts the existing tunnel fibers. Twelve detector modules are procured: ten cover the four cavity windows (two fibers each), the klystron output window and the circulator; two are spares. Two system modules and a test module form the external interface [R1]. The optical arc detectors at the cavity windows were never commissioned, and the arc front end in the Fast Interlock Chassis is largely disconnected [R3], [R6]; arcs at the cavities are caught only through the reflected-power and cavity-field faults [R5], so there the ARC4 adds protection rather than replacing a working function. Klystron and HVPS transformer arcs are caught by the arc triggers in the HVPS controller, which I1 and I2 do not touch (§4.4). Both LLRF9 units are installed in B132-13; their RF cabling is to be completed.

Interfaces.

- ARC4: the tunnel fibers, which arrive at the rear of the VXI crate and the Fast Interlock Chassis today [R2], move to the ARC4 once none is found to feed a live AIM arc channel (O3). The system-module outputs stay unconnected until I2, so until then the ARC4 is watched only at its front panel.
- LLRF9: taps and splitters on the legacy RF signals, and the reference from output RF3 of the VXI CLK module, as in 2021 [R8]. Every tap or splitter lowers a legacy signal, so the IQA calibrations and the protection thresholds are rechecked after each change; the 2021 couplers and splitters are reused where possible, the patch-panel record is updated, and the slave's circulator-load and waveguide-load forward signals need new taps [R8]. The same rules apply to the Waveform Buffer taps of I2.
- LLRF9 software: the vendor software of April 2026 has a channel map for each unit—Unit 1 the four probes, the klystron forward and the four cavity forward signals; Unit 2 the four cavity reflected, the klystron reflected, the circulator-load forward and the three waveguide-load forward signals—and also

functions that write to other IOCs: an HVPS loop, a tuner interface that writes the legacy stepper targets, and a station state machine [R39]. In I1 none of them may act on the station.

Worklist. Table 6.

Table 6: Worklist of I1.

Item	Work	Depends on
I1-H1	ARC4 chassis installed in B132 with its detector, system and test modules; its outputs left unconnected until I2	Doc 0 O5
I1-H2	Tunnel arc fibers moved from the rear of the VXI crate and the Fast Interlock Chassis to the ARC4, once none is found to feed a live AIM arc channel	O3, I1-T1
I1-H3	LLRF9 RF hookup as a controlled change: the 2021 couplers and splitters for the nine signals of Unit 1; new taps for the circulator-load and waveguide-load forward signals of Unit 2; the reference from CLK output RF3 to both units; the patch-panel record updated (A8)	—
I1-H4	Both LLRF9 units on the controls network	—
I1-S1	SW8: vendor IOC of April 2026 with the control channel map on Unit 1 and the monitor map on Unit 2 [R39]; calibration against the IQAs; screens, alarms, archiving and the Channel Access gateway	I1-H3
I1-S2	Monitor-only safeguards in the vendor IOC: the HVPS loop in its open mode, the tuner interfaces disabled, the station state machine not used, and no Channel Access write from the LLRF9 to B132-IOCRF or to any SRFTEST name [R39]	O7, I0-S3
I1-T1	Light test of every arc fiber end to end at the next tunnel access (none has been used in service); the ARC4 electronics exercised with its test module	—
I1-T2	Shadow comparison of the LLRF9 phases and amplitudes with the IQAs at several beam currents	I1-S1
I1-T3	IQA calibrations and protection thresholds rechecked after each tap	I1-H3
I1-T4	No legacy PV and no drive changes while the LLRF9 IOCs run, reboot and lose the network	I1-S2

Entry criteria. I1-S2 in place; the arc fibers checked against live AIM channels; the LLRF9 hookup planned as a controlled change with the threshold recheck.

Fallback. None needed in a trip path; a removed tap is reversed with the previous calibration constants.

6.3 I2: Interface Chassis and Waveform Buffer in Place of the Fast Interlock Chassis and the RF Local Display Panel

Scope. The production chassis, with the Interface Chassis and Waveform Buffer boards, replaces the Fast Interlock Chassis 340-308 and the RF Local Display Panel 340-310 in one down (Figure 3). The AIM, the RFP, the PLC-5, the tuner chassis and the HVPS controllers stay in service, so the Interface Chassis must work with each of them exactly as the two legacy units do. I2 is a replacement, not a series insertion: from I2 on, the Interface Chassis itself generates SCR ENABLE and KLYSTRON CROWBAR (rule 2 of §5.2). The ARC4, installed at I1, is read in from the start; its trip is enabled after a logged run.

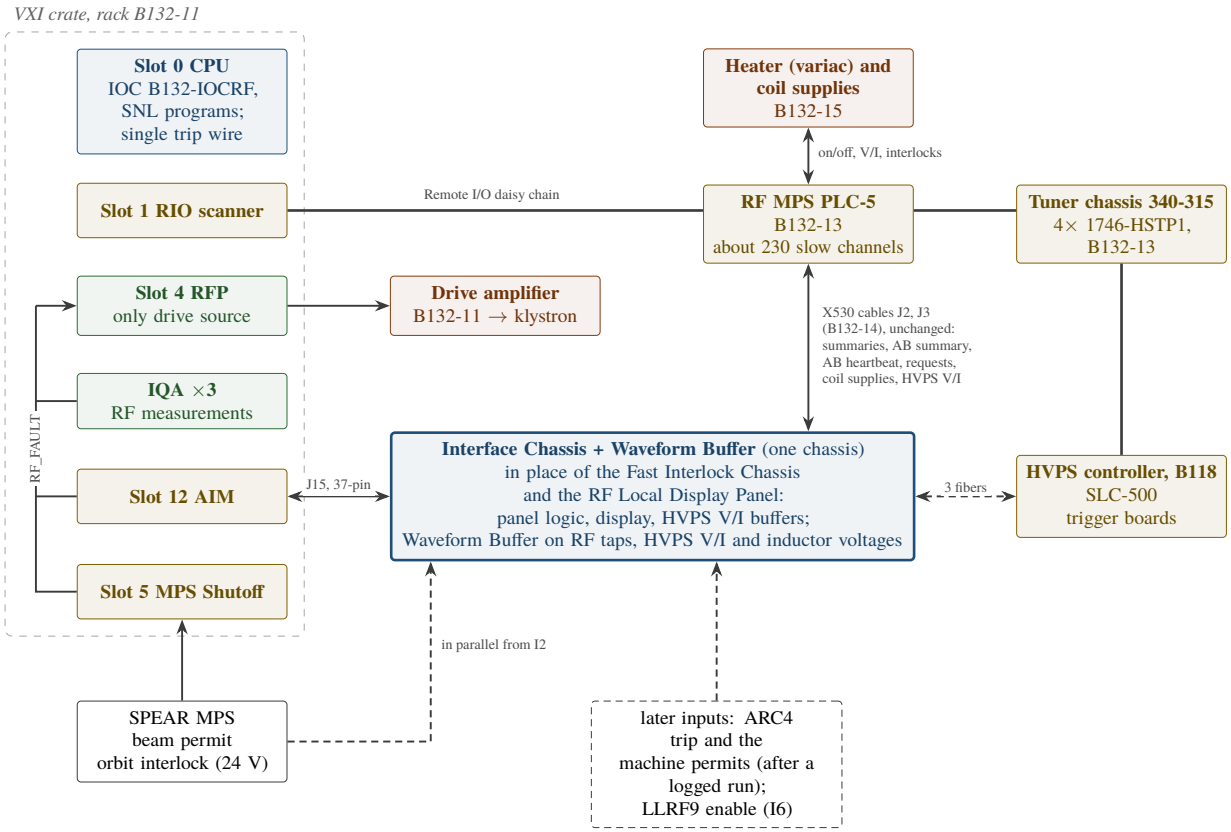


Figure 3: The station after I2. The Interface Chassis, with the Waveform Buffer in the same chassis, takes the place of the Fast Interlock Chassis and the RF Local Display Panel and presents their interfaces to the AIM (J15 and the 37-pin cable to J4), the PLC-5 (cross-connect X530) and the HVPS controller (three fibers); it takes the SPEAR MPS and orbit permits in parallel with Slot 5. Everything else is as in Figure 1, apart from the monitor-only ARC4 and LLRF9 units of I1. Dashed: fibers, and the inputs used later (the ARC4 trip and the machine permits after the logged run; the LLRF9 enable from I6 on).

Interfaces. Table 7 lists what the Interface Chassis must present at I2. These transition interfaces are not in J. Sebek's specification [R3], which describes the end state; Doc 0 includes them [R1]. Doc C sets the design requirements of the Interface Chassis and the Waveform Buffer for I2 and the end state: signals, levels, logic, timing and status reporting [R45].

Table 7: Interfaces of the Interface Chassis at I2.

Interface	Today	Requirement
AIM, cable J15	The Fast Interlock Chassis turns the AIM's six commands into fibers and returns the AB summary and heartbeat [R6]	Accept the J15 cable unchanged: receive the six commands at the levels the chassis receives them today; drive the AB summary fault (pin 10) and the AB heartbeat (pin 23) into the AIM's Schmitt-trigger inputs; leave the AIM's supply pins and trip-test lines unconnected. **[O3]**

Interface	Today	Requirement
AIM, 37-pin cable to J4	The Fast Interlock Chassis sends twelve arc signals and, on pin 34, the spare-fault line that carries Slot 5's J3 output, the AIM's External MPS channel; jumpers on the AIM enable the arc channels and External MPS (fault-enable word `0x71FB`, read 11 November 2026) [R27], [R28]	Accept the cable unchanged: hold pin 34 low while both machine permits are present, from the inputs themselves, so that External MPS appears as today; hold the arc lines at their no-arc level, so that the AIM's jumpers stay as they are; leave the AIM's 5 V on pins 35 to 37 unconnected; Slot 5's J3 cable labeled and left open [R45]
SPEAR MPS and orbit permits	Two 24 V permits into VXI Slot 5, which asserts RF_FAULT when either drops; the state machine trips on the AIM's latch of it [R2], [R3]	Paralleled into the Interface Chassis inputs; Slot 5 stays until I8 for the RF cut and the legacy software trip. Read from I2, the chassis records which permit dropped; once I2 is integrated they remove SCR ENABLE [R45] **[O6]**
PLC-5, cables J2 and J3 at X530	The panel exchanges 24 V lines with the PLC-5 and buffers the HVPS voltage and current (Appendix C)	Accept both cables unchanged **[O2]**; drive the request lines (station online, filament and focus, filament timer bypass, interlock reset); hold the online-mode line on, so that the station-online command from EPICS alone sets the PLC-5's online bit, and leave the local-mode line open [R45]; take the coil, waveguide-pressure and collector summaries into the logic and pass the AB summary and the AB heartbeat **[O4]** to the AIM; buffer the HVPS voltage and current into the PLC-5's analog inputs with unchanged scaling, because the collector power is computed from them. The lines that the panel only displays (HV and RF detected, filament, timeout, coil supplies on, the other area summaries) are not connected; the PLC-5 reports them to EPICS [R45]
Klystron forward power	A detector in the Fast Interlock Chassis: RF input J32 and output J31 on the rear, cabled to the PLC-5 input "RF PWR LEVEL" that its collector-power trip uses; the front monitor output J16 is not cabled [R1], [R6], [R41]	The same detector output on the J31 cable, with unchanged scaling
Coil supplies	PLC-5 outputs 032 and 033 carry the main-focus and bucking-coil summaries into the panel's logic: filament requested and supply voltage and current inside their limits, and for the main focus the supply's internal contact on input 025; the bucking-coil contact on input 026 is not used [R34], [R38]	Take both lines into the logic
HVPS controller, three fibers	The panel drives SCR ENABLE and KLYSTRON CROWBAR and receives STATUS [R4], [R34]	Same fibers and light conventions: lit SCR ENABLE enables the triggers, lit KLYSTRON CROWBAR means no crowbar, lit STATUS means the HVPS can run

Interface	Today	Requirement
Klystron arc	A Pearson current transformer in the termination tank feeds the klystron-arc trigger of the HVPS controller, which fires the crowbar [R5], [R25]	None at I2: the trip stays in the HVPS controller and reaches the Interface Chassis as a dark STATUS fiber (O3); I7 keeps the input
Transformer arc	The transformer-arc trigger enters the HVPS controller, which inhibits the triggers and fires the crowbar [R5]	As for the klystron arc
Beam abort	Beam abort on the panel's BNC J6, latched until reset; made for PEP-II and not cabled at SPEAR3 [R26], [R46]	None: the SPEAR MPS takes no status from the RF in hardware [R3], [R45]
Operators	Status display; station online/offline and local/remote key switches [R26]	Front-panel display only, with no key switches and no reset button: the station is set online or offline through `SRF1:STN:AIM:SOLENOID`, and reset, from EPICS; all states also in EPICS [R45]
Logic	GAL22V10 in the panel; its equations are not documented	Equations written from the required behavior of the system, with latching until reset as in the panel [R45]; compared with the recorded behavior of the panel (I0-T1), every difference documented and reviewed **[O1]**

The end-state inputs of the specification [R3]—LLRF9 status and enable, the RF MPS summary permit, heartbeat and external reset, SPEAR MPS and orbit permits, ARC4, power-monitor permit and expansion ports—are built in, each with a bypass or simulate jumper. The ARC4 input and the SPEAR MPS and orbit permits are read from I2 on and act after the logged run; the LLRF9 interlock output is read monitor-only from I3 on; the LLRF9 enable is used from I6 on. A command path from the Interface Chassis IOC that can replace J15 when the AIM goes at I8 (§6.9) is built in as well.

Worklist. Table 8.

Table 8: Worklist of I2.

Item	Work	Depends on
I2-H1	Production chassis with both boards: the transition interfaces of Table 7, the end-state inputs with bypass or simulate jumpers, the J15 command path for I8; isolated inputs and fail-safe outputs, so that loss of power or of an input removes the permits [R3]	O1 to O4, I0-H1
I2-H2	LLRF9 interface as the vendor specifies it, for the daisy chain of Doc 0 [R1]: Unit 2's interlock input, the head of the chain, driven high for enable with at least 7 mA at the level set at the factory (3.3 V, 5 V or 24 V); Unit 1's interlock output, the end of the chain, and the HVPS-interlock outputs of both units read as 4.75 V through 220 Ω, below 0.1 V when tripped [R40]	Doc 0 O1
I2-H3	A place in B132 that the J15 cable, the 37-pin cable to the AIM, the X530 cables, the three HVPS fibers, the SPEAR MPS and orbit permit wiring and the ARC4 outputs reach; otherwise new runs	—

Item	Work	Depends on
I2-H4	Waveform Buffer inputs: taps for its six RF channels (circulator-load reflected, station reference, waveguide loads 1 to 3 reflected and, proposed, the klystron drive power [R1]), installed as a controlled change; the HVPS voltage and current from the Interface Chassis buffers through high-impedance inputs; new cables from B118 for the transformer current and the SCR firing signal (Doc C O9). A Waveform Buffer fault reaches the Interface Chassis only through its comparator outputs	—
I2-H5	The Fast Interlock Chassis and the panel kept intact, their cables labeled, as the fallback and as spares until I8; the J15 cable and the 37-pin cable to the AIM moved to the Interface Chassis, so that the AIM's jumpers stay as they are; Slot 5's J3 cable labeled and left open	O3
I2-H6	The SPEAR MPS and orbit permits paralleled into the Interface Chassis inputs at their Slot 5 connection, within the current their sources can supply; Slot 5 stays until I8	O6
I2-S1	SW7: the Interface Chassis and Waveform Buffer IOC with the input and output states, the first-fault record, the ARC4 permit and fired-detector identity and the waveforms, in the alarm handler and the archiver	SW0
I2-S2	SW4: the ARC4 signals read out to the screens	I2-S1
I2-S3	Operator screens and procedures without the panel's key switches: the station online/offline control (`SRF1:STN:AIM:SOLENOID`) on the station screen; every procedure step that uses the panel keys, lockout and maintenance included, moved to EPICS or to the HVPS controller's PERMIT key [R45]	—
I2-T1	Bench: the logic against the equations of Doc C [R45] for every input combination, including latching and reset; every J15 and X530 line at its level, with simulators of the AIM and PLC-5 sides that reproduce the four PLC-5 lines into the logic and the heartbeat as the program forms it (§4.8); the fiber transmitters and receivers; power loss and input loss driving every output to its safe state; the first-fault record	O1, O2, O4
I2-T2	In place, without high voltage: every AIM command from EPICS through to the PLC-5; the AB summary and the heartbeat through to the AIM; online and offline from EPICS through to `SRF1:STN:LOCAL:ON` and SCR ENABLE; the scaling of the HVPS voltage and current at the PLC-5 inputs; the ARC4 inputs with its test module; each machine permit removed in turn, with the SPEAR MPS and orbit-interlock owners: Slot 5's RF cut and `SRF1:STN:VXI:LTCH` as before, External MPS on the AIM through the Interface Chassis, the permit named in its first-fault record; the AIM and PLC-5 records seeing the same bits, so that the legacy database needs no change	I2-H1
I2-T3	With high voltage, before beam: turn-on through the state machine; the responses to an HVPS trip, a klystron-arc and a transformer-arc trigger, a collector or waveguide-pressure summary, a VXI beam abort and a simulated coil-supply fault, each compared with the baseline	I0-T1
I2-T4	With beam: a run period with the ARC4 trips and the Waveform Buffer comparators logged but not enabled. The ARC4 input is enabled after a record without nuisance trips, which may be why the original optical system was never commissioned [R3]; the comparators after the Waveform Buffer's collector power has been compared with the PLC-5's; the SPEAR MPS and orbit permits once the record shows the Interface Chassis latching with Slot 5 on every machine-permit trip and at no other time	I1-T1, Doc 0 O5

Entry criteria. The production unit; O1 to O4 closed; the baseline of I0-T1; the failure-mode analysis; I2-S1 and I2-S3; I2-H4 installed as a controlled change; the fallback rehearsed. Before the ARC4 trip is enabled: I1-T1, Doc 0 O5 closed and the latch reset assigned to the ARC4 front panel or the Interface Chassis. Before the machine permits act: the record of I2-T4.

Fallback. Reinstall the Fast Interlock Chassis and the panel, and reconnect the J15, 37-pin and X530 cables, Slot 5's J3 cable and the fibers. The ARC4 trip and the machine permits alone are undone by masking their inputs.

6.4 I3: Tuner Control by LLRF9 Unit 1 and the Galil

Scope. The master LLRF9 unit (Unit 1) runs the four tuner loops and drives the Galil DMC-4143 directly through its built-in Galil driver [R1], while the RFP still drives the klystron. The legacy HVPS loop and state machine are unchanged, so I3 needs no change to the legacy SNL programs. The legacy tuner loops are switched off; the stepper chassis stays powered and connected as the fallback, which also keeps the tuner node alive on the RIO link.

Interfaces.

- Tuners: the motor cables move from the SS2000 drivers to the Galil. Unit 1 commands the tuner velocity through the Galil, which also reads the limit switches and has an analog input [R20], [R35]. The vendor's Galil database serves the DMC-21X3 and DMC-41X3 families, defines two axes and has no home or absolute move; it reads the Galil analog input as a potentiometer readback [R39]. The tuner motors have no encoders; the potentiometers give the only position readback and are used in no loop [R24]. They stay on the PLC-5 with the tuner temperatures and the bulk-supply channel (§4.7).
- EPICS: the legacy tuner loops are switched off (`SRF1:CAVTUNR:LOOP:CTRL`). That record and the loop status in its LOOP_OFF state are MAJOR alarms that roll up into the station severity summary, not into the trip summaries, so their severities or the alarm configuration are adjusted [R18], [R19].
- The LLRF9 interlock outputs are read into the Interface Chassis as monitor-only inputs, so that I6 starts with a record of them.
- A second actuator path exists in the vendor software: a SPEAR3 interface in which the LLRF9 loops write the target positions of the legacy steppers (`SRF1:CAVnTUNR:POSN:CTRL`, linked to SRFTEST names in the vendor files) once a second after the previous move has finished [R39]. It needs no cable change and can close the LLRF9 loops on the existing steppers before the Galil takes over, or keep them closed if the Galil fails.

Tuner control. The LLRF9 tuner loop is a PID controller with anti-windup on the phase between the probe and forward signals, computed every $0.1\,\mathrm{s}$, with a deadband, a loop sign, a load-angle offset and a minimum forward power below which it does not move the tuner; a probe-balancing loop can shift the offsets [R20], [R35], [R39]. The legacy loop runs every $2\,\mathrm{s}$ and moves a tuner only in TUNE and ON_CW above a minimum klystron forward power [R18], [R36]; a slower load-angle offset loop, when enabled, shifts the phase setpoints to balance the cavity voltages [R18], [R24]. Table 9 compares the two. SPEAR3 does not use the PARK state [R24], in which the state machine moves the tuners to park positions [R17], so no park move is needed.

Table 9: Tuner control in each operating condition.

Condition	Legacy tuner control	LLRF9 driving the Galil	Needed
Normal operation (ON or TUNE)	SNL loop every 2 s on the IQA load angle, with the setpoint shifted by the load-angle offset loop when it is enabled [R18], [R36]	PID every 0.1 s on the LLRF9's own probe and forward phases, with a fixed load-angle offset [R35]	Legacy loops off; LLRF9 offsets set from the legacy setpoints in shadow; any cavity balancing held at its last offsets until the coordinator or the vendor probe-balancing loop provides it
Turn-on	The state machine homes the tuners to their ON positions and waits 60 ticks (1 s) before enabling HV [R17], [R24]	No preset move in the LLRF9: the tuners start where they stopped, and the loop starts once the forward power exceeds its minimum	A home move on the Galil when the station is off, which J. Sebek asks for [R24] (SW2)
Commanded turn-off	Station OFF; the loop holds [R36]	Forward power below the minimum; the loop holds	None
Trip	As turn-off, after the RF cut	As turn-off; the few samples taken while the fields decay are limited by the deadband and the gains	Trip tests with beam confirm that the tuners do not move during the decay
LLRF9 IOC or unit down	—	The tuners hold; the detuning then follows beam current and temperature	Restore the IOC, use a spare unit [R37] or fall back to the legacy cables

Today every turn-on starts from the ON home positions, about 2.5 mm from where the loop then tunes the cavities at low power; in normal operation the tuners move about 0.2 mm [R24]. J. Sebek's operation note asks that the upgrade keep a defined home position, moved to when the station is off, and keep the tuner loops off until the cavity has a minimum of power [R24]. The LLRF9 does the second; SW2 adds the first on the Galil, so that a tuner does not start from wherever it stopped, for example from its full-current position after a trip.

Worklist. Table 10.

Table 10: Worklist of I3.

Item	Work	Depends on
I3-H1	The Galil cabled to the four tuner motors and their limit switches, and to Unit 1 over Ethernet	I3-T1
I3-H2	A position reference for the home move: the limit switches, or the potentiometers paralleled into the Galil analog inputs through high-impedance inputs, which leaves the PLC-5 readings unchanged	—
I3-S1	SW2: the vendor Galil configuration extended to four axes A to D, stepper with no encoder, in millimeters, with soft limits inside the legacy drive limits (18 mm and -29.5 mm in the archived database [R18]) and the mechanical restriction to ± 10 mm recorded in Doc L [R2]	I3-T1
I3-S2	SW2: a home move, not in the vendor software, made when the station turns off [R24]	I3-H2

Item	Work	Depends on
I3-S3	SW2: the vendor SPEAR3 tuner loops [R39]: probe and forward indices checked against the cabling of Unit 1; minimum forward power, deadband, gains and sign; the load-angle offsets set from the legacy setpoints in shadow	I1-T2
I3-S4	The legacy tuner loops off and the severities or alarm configuration of `SRF1:CAVTUNR:LOOP:CTRL` and the loop status adjusted [R18], [R19]	SW1
I3-S5	Optional first step and fallback: the vendor stepper interface re-pointed to the SRF1 names, with the legacy loops off	SW1, I0-S3
I3-T1	Bench: the Galil driver with the installed DMC-4143, whose screen is labeled for the DMC-21X3 [R20], and a spare motor: velocity commands, limit switches, stop, reconnection after a network loss	—
I3-T2	In place: the loops closed one cavity at a time, first at low power, with their detuning and reflected power compared with the legacy values	I3-S3
I3-T3	The home move and a turn-on from the home position	I3-S2
I3-T4	A trip with beam, checking that the tuners do not move while the fields decay	I3-T2
I3-T5	The stepper chassis, with its motors disconnected, raising no fault into the trip summaries; the tuners holding through an LLRF9 IOC reboot and a Galil disconnection	I3-H1

Entry criteria. I1, with the LLRF9 cross-calibrated against the IQAs (I1-T2); I3-S1 to I3-S4; the alarm changes tested.

Fallback. The motor cables back on the SS2000 drivers, the legacy step counters realigned to the potentiometers with the stop-and-init function [R24], and the legacy tuner loops on; or, with I3-S5, the LLRF9 loops on the legacy steppers.

6.5 I4: RF MPS

Scope. The ControlLogix replaces the PLC-5, both crates, in B132-13. Its program was written and tested on SPEAR3 without RF power [R1], [R37], and the system is online for software development with EPICS communication. The behavior it must reproduce is that of the decoded PLC-5 program (§4.8) as uploaded from the PLC-5 in service.

Interfaces.

- Field I/O through the 1771 field-wiring arms, reused by the Rockwell conversion kit [R1], [R37].
- The X530 lines to the Interface Chassis, as the PLC-5 drives them from I2 on: area summaries, AB summary and heartbeat, the filament, focus and bucking-coil control and the HVPS voltage and current inputs for the collector power [R7], [R34] (Table 29).
- EPICS: every channel of §4.7 under its legacy name, scaling and alarm limits, through legacy records re-pointed to the new IOC (§8.3).
- RIO: the PLC-5 node is first taken off the link with the IOC running, in a maintenance period before the window (I4-T1). The code predicts no trip, HWERROR on the station status and frozen PLC-5 values [R42]; the program predicts no change of the global sum or the heartbeat at the AIM [R38]. The IOC is not rebooted while the node is off. At I4 the trunk is spliced through to the tuner and HVPS nodes, and the legacy database change removes the rack-3 adapter, card and DCM records, or `config.ab` drops the node, so that the absent node raises no HWERROR.

Worklist. Table 11.

Table 11: Worklist of I4.

Item	Work	Depends on
I4-H1	The ControlLogix with the conversion kit in both crates of B132-13	—
I4-H2	The RIO trunk spliced through to the tuner and HVPS nodes and terminated as today	I4-T1
I4-H3	An analog output for the programming input of the heater supply, for I5 (§6.6)	—
I4-S1	Legacy-faithful program, checked rung by rung against the decoded PLC-5 program and the upload of the program in service: the global interlock sum with its online bypass, whose online-mode input the Interface Chassis holds on [R45]; latching and reset; the heartbeat and its module-error gating; the collector power and the HV-detected and RF-detected thresholds; the filament sequence; the focus and bucking-coil control and summaries; the cavity-vacuum time trips; the thermocouple conversions and temperature-rise trips **[O8]**	I0-T2
I4-S2	Legacy data image: the eight 64-word DCM pages as tag arrays, with the page number in word 3 (§8.4)	D5
I4-S3	Upgrade mode, switchable: Interface Chassis summary, heartbeat and reset, Interface Chassis status inputs, the heater analog ramp of I5, Waveform Buffer inputs	I4-S1
I4-S4	SW5: the RF MPS IOC generated from the legacy substitution files, the saved settings and the data image; the legacy records re-pointed through Channel Access, with a deliberate choice of the severity they take when the link is lost—today they freeze, while a maximize-severity link turns the station off	SW1, D3, D4
I4-T1	Node-removal test on the station, in a maintenance period, with the DCM record's LOCA field checked first: the PLC-5 and tuner nodes taken off the link with the IOC running, the AIM latches, the station status and the panel lamps watched, then the HVPS run at its minimum voltage through its own node, with the emergency-off buttons manned; the trunk kept continuous through to the HVPS node	SW1
I4-T2	The legacy database change rehearsed on the test IOC	I0-S2, I4-S4
I4-T3	The trip matrix (Doc L O2) on the ControlLogix, compared output by output with the PLC-5's	I0-T3, I4-S1
I4-T4	Every channel of §4.7 present with the same scaling and alarm limits; the data image compared word by word with recorded PLC-5 pages	I4-S2
I4-T5	The Interface Chassis accepting the summaries, the AB summary and the heartbeat; the heartbeat period against the I0 baseline	I0-T1
I4-T6	The collector power against the legacy value; the filament sequence on the variac: warm-up, the ramp window, high and low trips, bypass, blower run-on	I4-S1

Entry criteria. O8; I4-S1 to I4-S4; I4-T1 and I4-T2; the trip-matrix procedure and the I0-T3 baseline.

Fallback. The PLC-5 stays intact: the field-wiring arms moved back and the previous `config.ab` and `srf1.db` restored.

6.6 I5: Klystron Heater Supply under RF MPS Control

Scope. In the filament heater chassis in B132-15 the Genesys AC replaces the solid-state relay and the motor-driven variac; the chassis keeps its voltmeter and ammeter, the ammeter's current transformer, the output terminal block and connector J1 [R21]. With the RF MPS in place (I4), the supply goes straight to its target design: the RF MPS drives its analog programming input, and Ethernet carries monitoring and enable [R1]. The supply runs on a test stand with a load; its external 60 Hz sync circuit is in design [R1].

Interfaces. The filament request reaches the RF MPS from the AIM through the Interface Chassis [R6], [R7]. Today the PLC-5 drives two lines on J1, FILAMENT_ON to the solid-state relay and MOTOR-UP to the variac motor, and reads the voltage and current monitors there [R21]. The RF MPS instead enables the supply and ramps its analog programming input; the monitors keep feeding the RF MPS, so its heater interlock and limits are unchanged.

Operating point and limits. The heater runs at 68 V and 7.3 A at the chassis output [R1], within the chassis rating of 140 V and 1 kVA [R21]. Table 12 gives the PLC-5 trip limits, which the RF MPS keeps and which are read back into EPICS (`SRF1:FILAMENT:VOLT:ULIM`, `SRF1:FILAMENT:VOLT:LLIM` and the same for the current) [R13], and the live EPICS alarm limits. The archived substitution file holds different limits (95–128 V, 4.7–6.5 A) [R13]. They bracket neither today's operating point nor the 135 V that the chassis drawing gives as the expected value [R21], so any new database takes its limits from the live configuration.

Table 12: Live filament limits.

Channel	PLC-5 low trip	LOLO	LOW	HIGH	HIHI	PLC-5 high trip
Voltage (V)	62.4	63.8	66.5	72.1	76.2	76.2
Current (A)	5.24	6.72	7.01	7.59	7.88	9.77

Legacy sequence. The PLC-5 program runs the heater as follows, given with the values of the 2012 program; the program in service carries the 2021 changes (O8) [R38]. The RF MPS keeps all of it and, in upgrade mode, replaces MOTOR-UP by a ramp of the programming input to the same end point.

- The filament request starts a first timer (1 000 s) and switches FILAMENT_ON on, unless a high-trip latch is set.
- During the first timer MOTOR-UP is on while the voltage is below 100.0 V and the current below 5.20 A: the variac is raised to those constants, and raised again if it falls back, and is left where it is after the timer. The constants are in the ladder, not in EPICS; with today's high trip of 76.2 V they must have changed.
- A second timer (600 s) follows. The total, 1 600 s in 2012, has been 2 100 s since 2021. The timer bypass, held only while the filament is on, replaces the wait.
- A high trip on voltage or current switches FILAMENT_ON off at once and restarts the warm-up from zero after the reset; a low trip drops the filament summary. The low trips also latch at every cold start and need a reset once the heater is up.
- The filament summary is part of the global interlock sum. Without it the legacy database disables the HVPS contactor sequence, and the message logger opens the contactor on a fault [R12], [R44].
- The window-air blower runs while the filament is on and for 600 s after.

Worklist. Table 13.

Table 13: Worklist of I5.

Item	Work	Depends on
I5-H1	The Genesys AC in the heater chassis in place of the solid-state relay and the variac; the voltmeter, ammeter, current transformer, TB2 and J1 kept [R21]	I5-T1
I5-H2	The external 60 Hz sync circuit [R1]	—
I5-H3	The RF MPS analog output wired to the supply's programming input, with FILAMENT_ON (J1-A) as its hardware enable; the J1 monitors unchanged into the RF MPS	I4-H3
I5-H4	Ethernet from the supply to the heater IOC	—
I5-S1	RF MPS heater logic in upgrade mode: the ramp of the programming input in place of MOTOR-UP, ending at today's constants, with the timers, trips, bypass and blower of the legacy sequence	I4-S3, O8
I5-S2	SW3: the heater IOC over Ethernet for monitoring, supply status and the software enable [R1]; the hardware enable and the ramp stay with the RF MPS	SW0
I5-T1	On the test stand, closing Doc 0 O6: command set and IOC; regulation and waveform at 68 V and 7.3 A into a load like the klystron filament; lock to the 60 Hz sync; enable and inhibit wiring; monitor-output scaling against the RF MPS inputs; output held through a mains dip and through loss of the network	I5-H2
I5-T2	In place: the warm-up ramp; a high trip switching the supply off and restarting the warm-up; a low trip dropping the summary; the bypass; voltage and current at the operating point, inside the limits of Table 12	I5-S1
I5-T3	Klystron behavior at turn-on; the RF spectrum near the carrier compared with the variac baseline	I5-T2

Entry criteria. I4 with its heater ramp tested (I4-T6); I5-T1; the sync circuit; I5-S1 and I5-S2; a long down, because the cathode needs time to cool and re-warm.

Fallback. The variac chassis, kept in place.

6.7 I6: LLRF9 as Drive Source

Scope. The master unit (Unit 1), which has run the tuner loops since I3, takes over field control and drives the klystron through its outputs OUT0 and OUT2; the slave unit (Unit 2) carries the reflected and load signals and their interlocks [R1]. I6 goes in as soon as its software (SW9) is ready, in parallel with I4 and I5. The RFP stays as a hot standby on a transfer switch until I8, with the IQAs still measuring.

Interfaces.

- Drive: the drive-amplifier input in B132-11 becomes an interlocked RF transfer switch with its position read into EPICS, instead of a patch cable.
- Reference: output RF3 of the VXI CLK module, as in 2021 [R8], until the separate reference distribution of I8.
- Interlocks: the two units are daisy-chained, as decided in Doc 0 [R1]: the Interface Chassis enable drives Unit 2's interlock input, Unit 2's interlock output drives Unit 1's interlock input, and Unit 1's

interlock output returns to the Interface Chassis, which also reads the HVPS-interlock outputs of both units (LEMO; levels as in I2-H2 [R40]). A unit trips on its external input or any of its 17 internal sources; Unit 1 then turns off the drive and opens the absorptive switches in its output chains [R40], and a trip of Unit 2 reaches Unit 1 through the chain. The SPEAR MPS and orbit permits, wired into the Interface Chassis at I2 in parallel with Slot 5 (I2-H6), now also remove the LLRF9 enable.

- EPICS: the vendor station state machine and HVPS loop [R39], or the legacy HVPS loop fed from the LLRF9 (§8.6); exactly one loop writes `SRF1:HVPS:VOLT:CTRL`.

With the LLRF9 driving, RF_FAULT no longer removes the drive, and every fast RF-off path has to reach the LLRF9 another way (Table 14).

Table 14: Fast RF-off paths with the LLRF9 driving.

Fault source	Legacy fast action	Required with the LLRF9 driving
Reflected power, cavity field	IQA amplitude comparators assert RF_FAULT	The LLRF9's own interlocks: an overvoltage comparator on each RF input and eight slow-ADC window comparators [R40]
Arc (ARC4)	—	The Interface Chassis removes the LLRF9 enable and SCR ENABLE
SPEAR MPS beam permit, orbit interlock	Slot 5 asserts RF_FAULT; from I2 the Interface Chassis removes SCR ENABLE	The Interface Chassis also removes the LLRF9 enable
RF MPS trip	AB summary to the AIM; collector and waveguide-pressure summaries into the panel's logic	High voltage removed by the Interface Chassis logic as by the panel's, and the LLRF9 enable removed on the AB summary
HVPS controller trip, including its klystron-arc and transformer-arc triggers	High voltage removed; STATUS goes dark; the drive stays on until the state machine turns it off	STATUS also removes the LLRF9 enable [R3]
AIM and the other RF_FAULT sources in the crate	RF_FAULT	The state machine acts, unless an AIM output reaches the Interface Chassis (O3)

Worklist. Table 15. Two spare LLRF9 units exist [R37].

Table 15: Worklist of I6.

Item	Work	Depends on
I6-H1	The interlocked RF transfer switch at the drive-amplifier input in B132-11, its position read into EPICS	—
I6-H2	LLRF9 interlock wiring as a daisy chain: the Interface Chassis to Unit 2's interlock input, Unit 2's interlock output to Unit 1's input, Unit 1's interlock output and the HVPS-interlock outputs of both units to the Interface Chassis	I2, I2-H2
I6-H3	The SPEAR MPS and orbit permits, wired at I2, checked to remove the LLRF9 enable	I2-H6

Item	Work	Depends on
I6-S1	SW9: the vendor state machine configured for SPEAR3 (open-loop turn-on, tuning check, loop close, ramp, turn-off) and commanded by the minimal coordinator in step with the legacy state machine [R39]	O5, D11
I6-S2	SW9: one HVPS loop on the drive power—the vendor loop or the legacy loop fed from the LLRF9—and an interlock that keeps the other from writing the setpoint (§8.6)	D11
I6-S3	The vendor links re-pointed from the SRFTEST names to the SRF1 names: drive power, HVPS minimum and maximum, collector power and the HVPS setpoint [R39]	O7, I0-S3
I6-S4	SW9: the RFP-specific SNL functions disabled; the fault response (HVPS ramp-down and SCR disable, Interface Chassis reset)	O5
I6-T1	Bench: the LLRF9 interlock chain with the Interface Chassis, each internal source and the external input; the transfer-switch interlock	I6-H1, I6-H2
I6-T2	With the LLRF9 driving: the transfer at low power, following the switching, interlock and turn-on procedure of the 2021 tests [O7]; every path of Table 14; the hand-over of the HVPS loop, with only one loop writing the setpoint	I6-S1 to I6-S4
I6-T3	Stability, ripple rejection and synchrotron-line damping against acceptance criteria agreed before the window. In 2021 the LLRF9 rejected the ± 360 Hz and ± 720 Hz ripple by 16–21 dB against 24–27 dB for the legacy system, with the synchrotron line at 9 657 Hz and a quality factor of 16.7 against 8 932 Hz and 11.1 [R9]	I6-T2

Entry criteria. I2; O5, O6 and O7 closed; I6-S1 to I6-S4; I6-H1; the acceptance criteria; a record of the LLRF9 interlocks, kept from I3 on.

Fallback. The transfer switch back to the RFP and the legacy SNL configuration restored.

6.8 I7: New HVPS Controller and PPS Interface Box

Scope. The HVPS controller is replaced as a whole: a new Hoffman box, built around the CompactLogix, the FCOG1200 firing boards and the redesigned regulator, retires the existing box of the off-duty HVPS unit in B118 with its SLC-500, firing and regulator boards, trigger interconnect boards, power supplies and local control panel. In the same down and under one PPS certification the PPS Interface Box takes the PPS chains out of the PLC. The second unit follows at a later down, and both must be converted before I8 removes the RIO link; one unit is energized at a time, and the units exchange roles in scheduled downs, typically in April and August [R1]. The CompactLogix is online with EPICS communication and its ladder program is still being completed; the FCOG1200 boards are ordered and the regulator is being redesigned [R1].

Interfaces.

- Field cables: the existing cables to the HVPS power section, the contactor controller and the grounding tank land on the new box unchanged [R1], among them the klystron-arc coax from the Pearson current transformer and the transformer-arc trigger, which fire the crowbar in hardware [R5], [R25].
- HVPS fibers: the three fibers from the Interface Chassis with unchanged light conventions; the new box keeps the trigger inhibit of §4.4, which leaves B^+ and B^- firing so that the filter inductors discharge [R4], [R5].

- The HVPS voltage and current monitor outputs keep their scaling, because the Interface Chassis buffers them for the collector power and the Waveform Buffer.
- EPICS: all HVPS control stays in EPICS, with no Python layer, under the current PV names in B132-IOCRF: eleven records of the HVPS node plus 32 status bits (Appendix B) [R12]. While one unit is legacy and the other new, the IOC serves either through two database variants or a selector. The switch-over procedure moves six terminal blocks, the three fibers and the klystron-arc BNC in a switchbox in B118, the bus bars of the cable switch tank and the fan breakers in B514 [R31]; with one legacy and one new box, every line on those blocks has to be listed. [O9]
- RIO: the HVPS node removed and the chain re-terminated.
- PPS: today PPS 1 enters an SLC-500 input and the supply of the output that drives contactor K4, and PPS 2 enters another input while rung 0016 drives the Ross grounding-switch coil from a PLC output; chain 2 has no hardware protection against a PLC output stuck on [R25]. The PPS box drives K4 and the Ross coil directly, the Ross coil from PPS 2 alone where the legacy ladder needs both chains [R1], [R2]; this belongs in the PPS review. Two units need one box each, or a cable moved at the switch-over, which already breaks the PPS chain [R31].
- The local control panel (touch panel, PERMIT key switch, EMERGENCY OFF button with two contacts) is reproduced in the new box; the EMERGENCY OFF test of Doc L (O11) is part of the commissioning.

Worklist. Table 16.

Table 16: Worklist of I7.

Item	Work	Depends on
I7-H1	One new Hoffman box per HVPS unit: CompactLogix, FCOG1200 firing boards, redesigned regulator, the trigger inhibit of §4.4, and the local control panel with touch panel, PERMIT key switch and two-contact EMERGENCY OFF button	—
I7-H2	The PPS Interface Box (B. Morris): a Bud enclosure with four relays, status LEDs and one PPS-lockable connector, two permit and two readback channels, the standard design of the 6575 modulator and gallery systems [R1]; one per unit, or a cable moved at the switch-over	A14
I7-H3	The field cables landed on the new box unchanged, including the klystron-arc coax and the transformer-arc trigger; the three fibers; the voltage and current monitor outputs with unchanged scaling	O9
I7-H4	After both units: the HVPS node removed from the RIO link and the chain re-terminated	I7-T2
I7-S1	The CompactLogix program completed and reviewed against the SLC-500 program	—
I7-S2	SW6: the HVPS IOC behind the current names, and the selector or database variants while one unit is legacy and the other new; SW1 for the legacy database change	SW1
I7-T1	On Test Stand 18 in B44, upgraded for the new controller: the new box running the test-stand HVPS [R1]	I7-H1, I7-S1
I7-T2	At SPEAR3, the new box against the legacy unit: regulation and ripple; every HVPS interlock and crowbar source, including the klystron-arc and transformer-arc triggers; the fiber trio with the Interface Chassis; the HVPS loop in use (legacy or LLRF9) writing the same setpoint name	I7-T1

Item	Work	Depends on
I7-T3	PPS certification; the EMERGENCY OFF test of Doc L (O11)	A14

Entry criteria. The PPS steps of A14, ending in ADSO approval of the final design and documentation of the PPS box; the Radiation Safety Work Control Form; I7-H1 complete, bench-tested and run on Test Stand 18 (I7-T1); I7-S1 and I7-S2; O9 closed.

Fallback. Switch to the other HVPS unit, with its legacy Hoffman box and PPS wiring.

6.9 I8: Legacy Retirement

Scope. The VXI crate is removed with its CPU, RIO scanner, CLK module, RFP, IQAs, AIM and Slot 5, together with the tuner chassis, the RIO link and the spare Fast Interlock Chassis and panel. The IOC B132-IOCRF moves to new hardware with the full coordinator.

What changes.

- The AIM's commands to the Interface Chassis—station online, filament and focus, HVPS on and filament-timeout requests and the fault reset—come from the coordinator through the Interface Chassis IOC instead of J15.
- The SPEAR MPS and orbit permits reach only the Interface Chassis.
- The LLRF9 reference comes from a separate 476 MHz distribution.
- The transfer switch stays on the LLRF9, and the RFP is gone.
- Every legacy function of Table 18 has its new home.
- Links to legacy names move with the names: the vendor HVPS loop reads its drive power from an IQA today, so it needs another drive-power measurement [R39].

Worklist. Table 17.

Table 17: Worklist of I8.

Item	Work	Depends on
I8-H1	The separate 476 MHz reference distribution for the LLRF9 and the Waveform Buffer	—
I8-H2	New hardware for B132-IOCRF	—
I8-H3	A drive-power measurement for the HVPS loop that does not use an IQA: the proposed Waveform Buffer channel or an LLRF9 input	I2-H4
I8-H4	Removal of the VXI crate, the tuner chassis, the RIO link and the spare Fast Interlock Chassis and panel; as-built drawings of the new station [R37]	I8-T1
I8-S1	SW10: the full coordinator on the new hardware (§8.6)	D9
I8-S2	SW7: the command path from the Interface Chassis IOC in place of J15, among them the station-online command, the only online/offline control from I2 [R45]	I2-H1
I8-S3	Every Channel Access link to a retired legacy name re-pointed, among them the LLRF9's drive-power and collector-power links; legacy names that are not kept retired; screens, alarm tree and archiver moved	D3
I8-T1	SW10 tested on the new hardware with every subsystem in its final form	I1 to I7

Item	Work	Depends on
I8-T2	An end-to-end interlock test with fault injection on every layer	I8-T1
I8-T3	Full-power commissioning; operator training	I8-T2

Entry criteria. I1 to I7 complete, with both HVPS units on new boxes; a full run period of I6 without a fallback to the RFP; SW10 tested on the new hardware; the reference distribution; the Interface Chassis in hub mode and the RF MPS in upgrade mode; a home for every legacy function.

Fallback. None practical once the hardware is removed, hence the run period required at entry.

7 Legacy Functions and Their New Homes

Doc 0 assigns the legacy functions of Table 18, which have no one-to-one successor, to their new homes [R1]. With the panel and the Fast Interlock Chassis gone at I2, several pass through the Interface Chassis first.

Table 18: Legacy functions and their new homes.

#	Function	Legacy implementation	New home	Moves at
1	Beam abort to the SPEAR MPS	AIM force and reset by the state machine; the panel's logic; BNC J6, not cabled at SPEAR3 [R17], [R26]	None: the SPEAR MPS takes no status from the RF in hardware [R3], [R45]	—
2	Main focus and bucking-coil supply control and interlocks	PLC-5 through the panel [R7]	Interface Chassis passes the lines; later RF MPS outputs	I2; I4
3	Filament request and timer bypass	AIM, Fast Interlock Chassis, panel and PLC-5 [R6]	Interface Chassis passes the requests; later RF MPS and heater IOC	I2; I8
4	HV-detected and RF-detected lamps; online/offline and local/remote key switches	Lamps driven by the PLC-5; keys into PLC-5 inputs [R7]	Lamp states reported to EPICS by the PLC-5, later the RF MPS; online/offline from EPICS, with the online-mode line held on by the Interface Chassis; no local/remote key [R45]	I2
5	Buffered HVPS voltage and current for the collector power	Panel buffers [R7]	Interface Chassis buffers; later HVPS controller to RF MPS	I2; I7
6	HV turn-on gate	AIM "HVPS on" request into the panel's logic	Interface Chassis logic; the request from the coordinator after the AIM is gone	I2; I8
7	Automatic reset after a trip, with its retry count and forced-fault inhibit	State machine; AIM forced-fault latch, set by operator command [R17]	Coordinator	I8

#	Function	Legacy implementation	New home	Moves at
8	Reference power, crate and RIO health monitors	SNL and VXI records	Reference distribution monitor; IOC health records	I8
9	Fault capture and MATLAB fault tools	AIM, IQA and RFP history buffers; MATLAB tools	LLRF9 and Waveform Buffer post-mortem; ported tools	I6; I8
10	Readbacks for other subsystems	PLC-5 channels (§4.7)	RF MPS, with the same names	I4
11	476 MHz reference distribution	VXI CLK module	Separate reference distribution chassis	I8
12	Drive power for the HVPS loop	IQA measurement of the klystron drive [R16], also read by the vendor HVPS loop [R39]	Waveform Buffer drive-power channel or an LLRF9 input	I8

8 Software Plan

8.1 Coordinator and Principles

The coordinator is the IOC B132-IOCRF: a minimal release (SW9) on the VXI CPU for I6, which needs the legacy build environment (O5), and a full release (SW10) on new hardware for I8, each with one adapter per subsystem that talks to either the legacy or the new hardware, so that the order of I4 to I7 can change (§8.6). All HVPS functions stay in EPICS, with no Python layer. Principles:

1. **Minimum software per integration.** The legacy SNL programs stay in charge until I6, so every integration except I6 and I8 needs only its subsystem IOC and, at most, a legacy database change.
2. **Generate rather than write.** With a legacy data image in the new PLCs (§8.4), the legacy substitution files generate most of the RF MPS and HVPS databases; alarm limits come from the IOC's saved settings, because the archived files differ from the live values (§6.6).
3. **Standard modules and vendor software first**: the LLRF9's own IOC with its tuner loops, Galil driver, station state machine and HVPS loop [R39], a standard driver for the Genesys AC, and the EtherNet/IP driver for the ControlLogix and CompactLogix.

8.2 Work Packages

Every work package is led by the controls group, with technical support from the lead of the subsystem it serves, named in Table 19.

Table 19: Software work packages, led by the controls group.

WP	Content	Needed for	Relative size	Technical support
SW0	PV naming policy, legacy compatibility rules, one interface control document per integration	All	Small	Project lead
SW1	Owner for the legacy IOC; test IOC; procedure to edit and reload `srf1.db`; check of the SNL build	I3, I4, I6, I7	Small to medium	J. Sebek

WP	Content	Needed for	Relative size	Technical support
SW2	LLRF9 tuner loops driving the Galil: the vendor driver configuration extended to four axes, limits, home move and bench test; legacy tuner loops off and their alarms adjusted; optionally the vendor stepper interface	I3	Small	M. Dunning
SW3	Heater IOC (Genesys AC over Ethernet)	I5	Small	B. Morris
SW4	ARC4 permit and status, read out from the Interface Chassis IOC	I2	Small	D. Zhang
SW5	RF MPS IOC generated from the legacy substitution files, the saved settings and the legacy data image; legacy records re-pointed	I4	Medium	M. Larrus
SW6	HVPS IOC behind the current HVPS names in B132-IOCRF; selector in the legacy database for the legacy or new HVPS unit	I7	Small	M. Larrus
SW7	Interface Chassis and Waveform Buffer IOC: states, first fault, waveforms; later the command path that replaces J15	I2, I8	Medium	J. Sebek, D. Zhang, D. Scott
SW8	LLRF9 configuration, cross-calibration, screens, alarms and archiving; the monitor-only safeguards of I1 and the review of the vendor links to legacy PVs	I1, I6	Medium	J. Sebek, M. Dunning, D. Scott
SW9	Minimal coordinator in B132-IOCRF on the VXI CPU, commanding the vendor state machine and one HVPS loop (§8.6)	I6	Medium	J. Sebek
SW10	Full coordinator in B132-IOCRF on its new hardware, operator screens, post-mortem tools, ported MATLAB tools	I8	Large	J. Sebek

8.3 Adapting the Legacy IOC

When a legacy PLC is replaced, the legacy records that read it keep their names, alarm limits and summaries, and their data source changes from the RIO link to Channel Access links to the new IOC; the alarm tree and screens stay unchanged, and the new IOC keeps neutral names. Recreating the records in the new IOC under the legacy names would instead need every alarm limit and summary link reproduced exactly. Each legacy database change follows one procedure: edit; test on the test IOC; apply in a down with the previous `srf1.db` and `config.ab` kept; verify the alarm tree and the SNL behavior; record the configuration. A named owner for the legacy IOC and a test IOC are prerequisites for I3, I4, I6 and I7.

8.4 Legacy Data Image in the New PLCs

The PLC-5 and the SLC-500 exchange their data with EPICS through the RIO images of their DCMs, and the legacy records address these images by link, adapter, card and signal [R12], [R32]. If each new PLC program keeps tag arrays with exactly this layout, maintained by the ladder logic, the legacy records can be re-pointed one for one, the new IOC databases can be generated from the legacy substitution files, and old and new values can be compared word by word during commissioning. The cost is a small amount of ladder logic; the image is retired with the VXI crate.

For the PLC-5 the program gives the layout [R38]: eight pages of 64 words, each with its page number in word 3. Pages 0 and 1 hold the alarm limits of the 112 thermocouples; pages 2 and 3 their readings, with five temperature rises of the klystron (body, collector and window air) in place of five of its readings; page 4 the 32 raw analog inputs, the first four analog trip limits, the remaining warm-up time and the collector power with its limit; page 5 the other 60 analog trip limits; page 6 the fault latches, their copy frozen at the last global trip, the module communication bits and the summaries; page 7 only its number.

8.5 PV Naming Policy

Two IOCs serving one name give undefined Channel Access behavior. While a legacy record exists, its name belongs to B132-IOCRF; the HVPS keeps its current names, such as `SRF1:HVPS:VOLT:CTRL` and `SRF1:HVPS:VOLT:RBCK` [R12]; other new IOCs use prefixes that cannot collide with legacy names, such as `SRF1:MPS:` and `SRF1:WFBUF:` in Doc 0 [R1]. Exceptions are deliberate and recorded, and legacy names that are not kept are retired with the VXI crate.

8.6 Coordinator Scope

Minimal release (SW9), for I6.

- LLRF9 turn-on and turn-off: the vendor station state machine [R39], which resets the interlocks with a set number of retries, turns on open loop, waits until the tuning errors are in range, closes the field loop, ramps to the operating point and ramps down, configured for SPEAR3 and commanded in step with the legacy state machine.
- One HVPS supervisory loop on the LLRF9 drive power, chosen by D11 from the options below.
- Fault response: HVPS ramp-down and SCR disable, and the Interface Chassis reset. The automatic reset, with its retry count and forced-fault inhibit, stays in the legacy SNL.
- Adapters for the subsystems that are still legacy.

HVPS loop options for I6. Exactly one loop writes `SRF1:HVPS:VOLT:CTRL`; the others are held open, and the hand-over is tested (I6-T2).

1. **The vendor loop** [R39]. Every 0.5 s it compares the klystron drive power with a setpoint of 28 W, or 29 W above a klystron forward power of 600 kW, and outside a deadband of 0.1 W lowers the HVPS voltage by 0.05 kV per watt of shortfall, raising it for an excess, at most 0.5 kV per step, between the legacy minimum and maximum. Above a collector power of $1\,000$ kW from the PLC it only lowers the voltage; when the LLRF9 interlock trips it sets the minimum once. It runs only with its mode set to closed and the field loop closed, and it ships open. Its inputs are the IQA drive power and legacy records, under SRFTEST names, and its output is a record inside the LLRF9 IOC, so all are re-pointed (I6-S3). It has no states, status codes or tolerance checks of the legacy kind.
2. **The legacy loop, database only.** J. Sebek proposes keeping the legacy HVPS loop while the LLRF9 drives. The loop regulates on database records, such as the HVPS step `SRF1:KLYSDRIVFRWD:HVPS:DELTA` computed from the drive power and its setpoint, and checks the RFP's direct-loop flag and severity, `SRF1:STN:RFP:MODU.DLE` and `SRF1:STN:RFP:MODU.SEVR` [R16]. Fed from the LLRF9, or held in their operating state while the RFP stands by, these let the loop keep regulating with no SNL change. The flag also acts on the RFP and the DAC loop, so the option is proven on the test IOC first.
3. **The legacy loop, changed in SNL** so that it no longer depends on the RFP, keeping its states, status codes and tolerance checks; this needs the SNL build (O5).

With option 1 or 2, I6 needs the SNL build only for other SNL changes, and SW9 shrinks to commanding the vendor state machine and the fault response. All three read the klystron drive power from an IQA, which goes at I8 (Table 18, row 12).

Full release (SW10), for I8. On the new hardware of B132-IOCRF: station state machine, the commands to the Interface Chassis that the AIM sends today, automatic recovery with its retry count and forced-fault inhibit, station-level calibration, the load-angle offset loop that balances the cavity voltages through the LLRF9 offsets [R20], [R24], the tuner home move, fault manager and post-mortem from the LLRF9 and Waveform Buffer data, operator and expert screens, alarm and archiver configuration, and ported MATLAB tools.

9 Deferral and Contingency

9.1 If an Integration Slips

Table 20: Effect of a slipping integration.

If this slips	Still possible	Blocked	Mitigation
ARC4 (I1)	All	None	The Interface Chassis input stays masked
LLRF9 measurement (I1)	I2, I4, I5, I7	I3, I6, I8	The Interface Chassis and the Waveform Buffer still add protection and diagnostics
Interface Chassis production (I2)	I1, I3, I4, I5, I7	The ARC4 trip; I6; I8	An interim ARC4 trip point in the legacy chain, chosen from the Fast Interlock Chassis and AIM drawings [R27], [R28]; for the LLRF9, the existing LLRF Window Interlock Status chassis, which turned a legacy permit into an LLRF9 enable in 2021 (O7)
Galil (I3)	All others; I6 with the legacy tuner loops on IQA phases	I8	None
RF MPS (I4)	All except I5 in its target form and I8	I5 under RF MPS control; I8	The heater goes in with the legacy enable and monitors
Heater (I5)	All	None, if the RF MPS keeps the legacy heater I/O	None
EPICS-side software	I1 with the vendor LLRF9 IOC	Every later integration	Generated databases, standard drivers and vendor software, minimal coordinator first
Legacy build environment (O5)	I1 to I5 and I7, with database-only changes; I6 with the vendor state machine and an HVPS loop that needs no SNL change	I8; I6 if it needs an SNL change	Find or rebuild the environment early; expert-run LLRF9 tests as in 2021; options 1 and 2 of §8.6 for I6
New HVPS controller (I7)	All other integrations	I7, I8	The legacy Hoffman boxes stay in service
PPS approval	All other integrations	I7, I8	Complete the documentation early; the new HVPS PLC must not carry the PPS chains, so I7 waits

9.2 If Legacy Hardware Fails During the Transition

Table 21: Contingency paths for legacy failures.

Failure	Effect	Contingency path
Fast Interlock Chassis or RF Local Display Panel, before I2	Station down	Interface Chassis with the interfaces of I2
VXI CPU	Station down, including B132-IOCRF	A spare CPU; moving B132-IOCRF early needs the I8 prerequisites
AIM, after I2	Commands to the Interface Chassis lost; station down	The command path from the Interface Chassis IOC (§6.3), if built in from the start
RFP or IQA	Station down	LLRF9 drive configuration (I6 prerequisites), with an interim permit adapter if the Interface Chassis is not yet installed
CLK module	Station down; no LLRF9 reference	The separate reference distribution of I8
RIO scanner	PLC data and HVPS setpoint lost	RF MPS and HVPS controller on Ethernet (I4 and I7 prerequisites)
PLC-5	Station down	ControlLogix with the legacy-faithful program and IOC (I4 prerequisites)
Hoffman box of the active HVPS unit	Unit down	Other unit; a new box once the I7 prerequisites exist
Tuner controller or driver	Tuning lost	LLRF9 driving the Galil (I3 prerequisites)
Variac or its motor	Heater control lost	Genesys AC (I5 prerequisites)

Preparing each integration's prerequisites therefore also builds an emergency path for the legacy element it replaces, one more reason to keep the new subsystems compatible with the legacy station.

10 Risks

Table 22: Integration risks.

#	Risk	Consequence	Mitigation
1	No owner or build environment for the legacy IOC	The minimal coordinator and every legacy database change blocked, and with them I3, I4, I6 and I7	Name an owner; set up a test IOC; find the vxWorks build environment early (O5); prefer database-only changes
2	The Interface Chassis logic misses or changes a function of the panel	The Interface Chassis behaves differently from the legacy chain	Equations written from the required behavior and reviewed (O1); baseline test in I0; equivalence test before I2; if needed, a truth-table test of the spare panel board
3	Transition interfaces missing from the production Interface Chassis	I2 slips, and with it the ARC4 trip, I6 and I8	Add them at the production review (D6)

#	Risk	Consequence	Mitigation
4	The Interface Chassis fails after I2	Station down	The Fast Interlock Chassis and the panel kept as spares until I8; spare boards
5	Software capacity	Hardware waits for software	Generated databases, standard drivers and vendor software, two coordinator releases
6	Hidden behavior of the PLC-5 program	Unexpected trips or a missing interlock after I4	The program decoded (§4.8) and compared with the one in service (O8); legacy-faithful mode, legacy data image, full trip matrix
7	RF hookup changes legacy calibrations	Protection thresholds move	Controlled change with recheck of thresholds
8	Drive-source mix-up	Klystron driven by an unintended source	Interlocked transfer switch with readback
9	Loss of the fast RF cut with the LLRF9 driving	Delayed RF removal after a SPEAR MPS or orbit trip	SPEAR MPS and orbit permits in the Interface Chassis from I2 (O6); every path of Table 14 tested
10	PV name collisions	Undefined Channel Access behavior	Naming policy (SW0)
11	RIO chain broken during I4 or I7	Loss of HVPS and tuner data	Splice prepared and tested; node terminals kept connected
12	PPS approval delayed	I7 blocked, because the HVPS controller and the PPS box go in together	Complete the documentation and submit it for ADSO approval early
13	Same people needed for several integrations	Integrations slip together	Windows planned by person (rule 4 of §5.2)
14	Nuisance trips from the ARC4	Lost beam time; trip disabled	Monitor-only record before enabling
15	A solid-state amplifier project starts during the migration	Priorities and people diverted	Keep the integrations independent; the Interface Chassis and coordinator design do not depend on the RF source [R22]
16	Legacy functions discovered late	VXI retirement delayed	Table 18 reviewed with operations and the SPEAR MPS group
17	Tuner control depends on the LLRF9 IOC	No tuning while it is down	Legacy tuner hardware kept connected as fallback; spare LLRF9 units [R37]; IOC health alarm
18	LLRF9 vendor functions that write to other IOCs: the HVPS loop, the stepper interface and the state machine [R39]	An unintended HVPS or tuner move, or two loops writing the HVPS setpoint	Vendor software reviewed (I0-S3); monitor-only safeguards in I1 (I1-S2); links re-pointed only in I3 or I6, with one writer of `SRF1:HVPS:VOLT:CTRL`

11 Decisions

Table 23: Decisions.

ID	Decision	Recommendation or outcome	Needed by
D1	Integration sequence	Decided: I1 to I8 as in Table 4	Decided
D2	Tuner control	Decided: the LLRF9 tuner loops drive the Galil directly; no separate Galil IOC	Decided
D3	PV naming policy	Legacy names stay with B132-IOCRF; the HVPS keeps its names; new IOCs use non-colliding prefixes (§8.5)	Before SW5
D4	RF MPS EPICS path	Re-pointed legacy records with a new IOC (§8.3)	Before SW5
D5	Legacy data image	In both new PLC programs	Before SW5 and SW6
D6	Interface Chassis production scope	The transition interfaces of Table 7, the end-state inputs and the command path that replaces J15 at I8	Production review (I0)
D7	Heater control	Decided: under RF MPS control from I5; the legacy enable only if I4 slips	Decided
D8	HVPS controller scope	Decided: a complete new Hoffman box for each HVPS unit retires the existing one	Decided
D9	Coordinator releases	Minimal (I6) and full (I8), in B132-IOCRF, with per-subsystem adapters	Now
D10	Legacy IOC owner	Proposed: D. Scott (controls group); at least technical support on SW7 and SW8	Now
D11	HVPS loop while the LLRF9 drives	Open: the vendor loop or the legacy loop with database-only changes, chosen after the review of I0-S3 and a test of the second on the test IOC; the SNL change only if neither serves (§8.6)	Before SW9

12 Actions

Table 24: Actions.

#	Action	Suggested lead	For
A1	Record decisions D1 to D11	Project lead	I0
A2	Name a legacy IOC owner; set up a test IOC; check the SNL build (O5)	D. Scott (controls group)	I0
A3	Record the baseline test of the legacy protection chain	J. Sebek	I0
A4	Review the Interface Chassis logic equations (O1), check the X530 lines (O2) and confirm the Fast Interlock Chassis functions (O3)	J. Sebek, D. Zhang	I1, I2
A5	Measure the legacy heartbeat (O4)	J. Sebek	I2
A6	Hold the production review of the Interface Chassis and Waveform Buffer with the requirements of §6.3 and Doc C [R45]	J. Sebek, D. Zhang	I2
A7	Define the ARC4 output interface (Doc 0 O5) and plan the fiber test	J. Sebek, D. Zhang	I1

#	Action	Suggested lead	For
A8	Plan the RF hookup as a controlled change, including the threshold recheck	J. Sebek	I1
A9	Trace the Slot 5 permit loop (O6)	J. Sebek	I2
A10	Recover the 2021 LLRF9 configuration (O7)	J. Sebek	I6
A11	Configure the LLRF9 tuner loops and the Galil driver; bench-test with the DMC-4143; run the shadow comparison	M. Dunning	I1, I3
A12	Complete the heater test-stand acceptance and the sync circuit	B. Morris	I5
A13	Run the PLC-5 trip matrix	S. Condamoor	I4
A14	Approve the PPS Interface Box in four steps: (a) a proposal to separate the PPS wires and connectors, from J. Sebek, B. Morris and M. Larrus; (b) its confirmation with J. Schroder and P. Schuh; (c) a review with S. Fleming's safety group; (d) the final design and documentation submitted for ADSO approval	B. Morris	I7
A15	Write the HVPS unit transfer list and the switch-over procedure (O9)	M. Larrus	I7
A16	Upload the PLC-5 program in service and compare it with the decoded 2012 program (O8)	S. Condamoor	I4
A17	Review the LLRF9 vendor software of April 2026 on a spare unit: its links to legacy PVs, HVPS loop, tuner interfaces and state machine	J. Sebek, M. Dunning	I1

A Source Index

Ref	Source	Location
[R1]	Doc 0, SPEAR3 LLRF Upgrade System Design Report, R4	`Designs/tex/doc0-system-design/`
[R2]	Doc L, Legacy System Architecture, R1	`Designs/tex/docL-legacy-architecture/`
[R3]	J. Sebek, LLRF Interface Chassis specification	`llrf/architecture/llrfInterfaceChassis.docx`
[R4]	J. Sebek, fiber-optic connections between the RF system controllers (HVPS controller side)	`hvps/architecture/designNotes/controllerFiberOpticConnections.docx`
[R5]	J. Sebek, RF system MPS requirements	`hvps/architecture/designNotes/RFSystemMPSRequirements.docx`
[R6]	J. Sebek, LLRF documentation notes, R2 (RIO topology, fibers, RF Local Display Panel, Fast Interlock Chassis)	`llrf/documentation/LLRFDocumentationNotesR2.docx`
[R7]	RF Local Display Panel to cross-connect mapping (SD-340-311-01 against WD-340-330-02)	`llrf/documentation/LocalPanelToXConnectMapping.xlsx`
[R8]	B132-11 patch panel configuration and coupler losses	`llrf/calibrations/b132R11PatchPanel.xlsx`
[R9]	J. Sebek, comparison of the LLRF9 with the legacy LLRF system (2021)	`llrf/tests/llrf9Tests.tex`
[R10]	Legacy IOC startup script, head revision 1.18	`spear-rf-code-legacy/iocBoot/`

Ref	Source	Location
[R11]	Legacy RIO scanner configuration (`config.ab,v`)	`spear-rf-code-legacy/iocBoot/`
[R12]	Legacy HVPS database (`rf_hvps.db,v`)	`spear-rf-code-legacy/rfApp/`
[R13]	Legacy analog channel template and substitutions (`rf_analog.db`, `rf_analog_All`, `rf_analog_4CV`)	`spear-rf-code-legacy/rfApp/`
[R14]	Legacy temperature, interlock and digital substitutions	`spear-rf-code-legacy/rfApp/`
[R15]	Legacy tuner loop PV definitions (`rf_tuner_loop_pvs.h,v`)	`spear-rf-code-legacy/rfApp/`
[R16]	Legacy HVPS loop PV definitions (`rf_hvps_loop_pvs.h,v`)	`spear-rf-code-legacy/rfApp/`
[R17]	Legacy station state machine (`rf_states.st,v`)	`spear-rf-code-legacy/rfApp/`
[R18]	Legacy cavity database with the tuner records (`rf_cav.db,v`, `rf_stn_cav.db,v`)	`spear-rf-code-legacy/rfApp/`
[R19]	Legacy cavity and station summaries (`rf_sumy_cav.db,v`, `rf_sumy_stn.db,v`)	`spear-rf-code-legacy/rfApp/`
[R20]	LLRF9 tuner loop and Galil screens (`tuner.edl`, `motor_galil.edl`)	`llrf/llrf9/iGp/dl_llrf/`
[R21]	Klystron filament heater control chassis drawing	`llrf/documentation/filamentHeater/sd3403110002.pdf`
[R22]	SSA upgrade summary, v11 (Test Stand 18 status, proposed SSA project)	`SSA_upgrade/`
[R23]	J. Sebek, interfaces between the RF system controllers	`hvps/architecture/designNotes/interfacesBetweenRFSystemControllers.docx`
[R24]	J. Sebek, SPEAR3 RF station operation	`llrf/documentation/LLRFOperation_jims.docx`
[R25]	Hoffman box PPS wiring	`pps/HoffmanBoxPPSWiring.docx`
[R26]	RF Local Display Panel drawings: drawing list DL-340-310-00, front and rear panels SI-340-310-05 and -02, schematic SD-340-311-01 (“PEP-II RF System Local Panel”)	`llrf/documentation/localPanel/`
[R27]	Fast Interlock Chassis schematics SD-340-308-01 and -02	`llrf/documentation/legacyInterfaceModules/`
[R28]	AIM schematic SD-340-309-01	`llrf/documentation/legacyInterfaceModules/sd3403090102.pdf`
[R29]	Wiring diagram WD-340-330-29 (control fiber-optic cables of the RF Local Display Panel)	`llrf/documentation/mpsWiringDiagrams/wd3403302900.pdf`
[R30]	Wiring diagram WD-340-330-02 (RF Local Display Panel cross-connects)	`llrf/documentation/mpsWiringDiagrams/wd3403300200.pdf`
[R31]	HVPS switch-over procedure, R0	`hvps/documentation/procedures/spearRfHvpsSwitchProcedureR0.docx`
[R32]	Legacy RIO adapter substitutions (`rf_ab_4CV.substitutions,v`)	`spear-rf-code-legacy/rfApp/`
[R33]	Galil test records (2024–2025) and commissioning notes	`llrf/tuners/galil/`

Ref	Source	Location
[R34]	J. Sebek, fiber-optic connections between the LLRF system and the HVPS controller, Rev 3	`llrf/documentation/fiberOpticCableSignalControlRev3.docx`
[R35]	Dimtel, LLRF9 manual (SPEAR3 screens)	`llrf/llrf9/llrf9_manual_s3_screen.pdf`
[R36]	Legacy tuner loop program (`rf_tuner_loop.st`,v)	`spear-rf-code-legacy/rfApp/`
[R37]	J. Sebek, LLRF upgrade task list, Rev 3	`llrf/documentation/LLRFUpgradeTaskListRev3.docx`
[R38]	RF MPS PLC-5 program: four RSLogix 5 projects (undated, November 2011, February 2012), the decoder and the decoded listings	`llrf/allenBradley/`
[R39]	Dimtel, LLRF9 software of 29 April 2026: EPICS IOC with the SPEAR3 channel maps, tuner, Galil and stepper interfaces, HVPS loop and station state machine; FPGA image; IOC installer	`llrf/llrf9/llrf9_2026/`
[R40]	Dimtel, LLRF9 manual, revision 1.6 (12 November 2025)	`llrf/llrf9/llrf9_2026/llrf9_manual_screen.pdf`
[R41]	Wiring diagrams WD-340-330-21, -22, -23 and -25 (PLC-5 analog inputs 1 and 2, digital input 1, digital output 1)	`llrf/documentation/mpsWiringDiagrams/`
[R42]	Legacy RIO scanner driver and DCM record and device support	`spear-rf-code-legacy/allenBradley/`
[R43]	Legacy build configuration and release notes (`RELEASE`,v, `RELEASE_NOTES`,v)	`spear-rf-code-legacy/`
[R44]	Legacy message logger program (`rf_msgs.st`,v)	`spear-rf-code-legacy/rfApp/`
[R45]	Doc C, Interface Chassis and Waveform Buffer Design Requirements, R0	`Designs/tex/docC-interface-chassis/`
[R46]	PEP-II RF station block diagram BD-340-330-00	`llrf/documentation/legacyArchitecture/bd3403300000.pdf`

B Legacy EPICS Interface Details

All record names carry the prefix `SRF1:`.

B.1 HVPS Records on the HVPS RIO Node

Table 25: HVPS records on the HVPS RIO node (logical rack 1) [R12].

Record	Type	Function	Address
`HVPS:CURR`	ai	Measured current	A1 C8
`HVPS:VOLT`	ai	Measured voltage	A1 C6
`HVPS:VOLT:RBCK`	ai	Voltage setpoint readback	A1 C4
`HVPS:VOLT:ULIM`	ai	Voltage limit readback	A1 C10

Record	Type	Function	Address
HVPSAC:CURR	ai	AC line current	A1 C2
HVPS:VOLT:CTRL	ao	Voltage setpoint, written by the HVPS loop	A1 C2
HVPS:VOLT:MAX	ao	Voltage limit	A1 C4
HVPS:RESET:CTRL	bo	Reset	A1 C10
HVPSCONTACT:CLOSE:CTRL	bo	Contactor close	A1 C6
HVPSSCR:ON:CTRL	bo	SCR enable, set by the state machine	A1 C8
HVPSDCM:T0:STAT	mbbi	DCM table status	A1 C0

In addition, 32 status bits of the HVPS controller are read from the HVPS node through the digital templates [R14], and the HVPS oil temperature comes from the PLC-5.

B.2 Template Instances Fed by the PLC-5

Table 26 counts the template instances that take their data from the PLC-5, from the head revisions of the substitution files loaded by the IOC. Each instance holds several records (value, alarm latch, summary links). A few further records in the klystron, station and HVPS databases also read the PLC-5.

Table 26: Legacy template instances fed by the PLC-5 [R13], [R14].

Template	Content	Instances
rf_temp	Temperatures	111
rf_interlock	Interlock bits	41
rf_analog	Analog channels with PLC-5 trip-limit readbacks, including tuner positions, filament, focus, tuner supply and the HVPS voltage and current seen by the PLC	20
rf_analog_log	Logarithmic analog channels (vacuum, ion pumps)	8
rf_sumy_plc	PLC summaries	24
rf_ab_module	PLC module status	16
rf_digital_plc	PLC digital status	12
Total		232

B.3 SNL Bindings Relevant to the Migration

Table 27: SNL bindings relevant to the migration.

Program	PVs	Use	Effect of the migration
HVPS loop	STN:RFP:MODU.DLE, STN:RFP:MODU.SEVR	Regulation mode; RFP health	With the RFP idle the loop regulates on the wrong variable or alarms [R16], unless they are fed from the LLRF9 (§8.6)

Program	PVs	Use	Effect of the migration
HVPS loop	HVPS:VOLT:CTRL, HVPS:VOLT	Setpoint and readback	Re-pointed to the HVPS IOC [R16]
HVPS loop	KLYSOUTFRWD:POWER, KLYSDRIVFRWD:POWER:ERR, STN:VOLT	IQA inputs	Must stay valid while the legacy HVPS loop runs [R16]
Tuner loop	CAVnTUNR:STEP:MOTOR fields RBV, DRVH, DRVL, DMOV, RDBD	Motor readback, limits, done flag, deadband	Unused with the LLRF9 driving the Galil; kept for the fallback [R15], [R18]
Tuner loop	CAVnTUNR:POSN:CTRL	Target position	Unused with the legacy loops off [R15]
Tuner loop	CAVnTUNR:POSN	Potentiometer readback from the PLC-5	Kept through the RF MPS cutover [R15]
Tuner loop	KLYSOUTFRWD:POWER:MIN, CAVnLOAD:ANGLE:ERR	Minimum power; phase error	From the IQAs [R15]
State machine	CAVTUNR:LOOPON:RESET, CAVTUNR:LOOPPARK:RESET	Tuner home and park, then a fixed wait	No effect on the tuners driven by the LLRF9; the Galil home move of SW2 replaces it [R17]
State machine	STN:AIM:FRCBMABT, STN:AIM:MODU.RBA	Force and reset of the beam abort	Not carried over: the Interface Chassis has no beam-abort output [R17], [R45]
Message logger	HVPSCONTACT:CLOSE:CTRL, filament latches	Opens the HVPS contactor on any filament fault	The filament latches keep their legacy names under the RF MPS [R44]
LLRF9 HVPS loop (vendor)	KLYSDRIVFRWD:POWER, HVPS:VOLT:MIN, HVPS:VOLTULIM:CTRL, KLYSCOLLPLC:POWER, prefix SRFTEST	Loop inputs	Re-pointed to SRF1 at I6; drive power replaced at I8 [R39]
LLRF9 stepper interface (vendor)	CAVnTUNR:POSN:CTRL, CAVnTUNR:STEP:MOTOR.RBV, prefix SRFTEST	Stepper target and readback	Disabled in I1; optional in I3 [R39]

C RF Local Display Panel I/O

The RF Local Display Panel 340-310 sits in rack B132-12 and connects to cross-connect block X530 in rack B132-14 through two 25-pin connectors [R6], [R7], [R26], [R30]. The Interface Chassis reproduces this interface at I2 (Table 7). Its eleven fibers are listed in Table 28 [R6], [R26], [R29].

Table 28: Fibers of the RF Local Display Panel.

Panel device	Panel label	Light means	Other end
U10	VXI beam abort	No abort	Fast Interlock Chassis (BEAM_ABORT)
U11	HVPS beam abort	No abort	HVPS controller (STATUS)
U12	HVPS on request	Request	Fast Interlock Chassis (HVPS_ON)

Panel device	Panel label	Light means	Other end
U13	Station online request	Request	Fast Interlock Chassis (SOLENOID_ON)
U14	Filament and focus request	Request	Fast Interlock Chassis (FILAMENT_ON)
U15	Filament timeout bypass	Bypass	Fast Interlock Chassis (FILAMENT_TIMEOUT)
U16	Reset	Reset	Fast Interlock Chassis (FAULT_RESET)
U19	HVPS triggers enable	HVPS on	HVPS controller (SCR ENABLE)
U20	HVPS crowbar	No crowbar	HVPS controller (KLYSTRON CROWBAR)
U21	AB heartbeat	Pulsing	Fast Interlock Chassis (HEARTBEAT)
U22	AB summary	No fault	Fast Interlock Chassis (A/B SUMMARY)

The fiber-cable diagram labels the Fast Interlock Chassis ends of the U13 and U14 fibers FILAMENT and SOLENOID [R29], and J. Sebek's notes follow it [R6]. Those two labels are swapped: the panel drives the solenoid-request line from U13 and the filament-request line from U14 [R26], and the legacy software treats SOLENOID as the station online/offline command and FILAMENT as the filament [R44]. The copy of the drawing in the repository carries a note.

The labels below are those of the panel schematic SD-340-311-01 with the cross-connect labels of WD-340-330-02; the source does not state signal directions. The labels suggest that the "Req" lines carry requests from the panel to the PLC-5 and the "Ctrl" lines carry the PLC-5's commands. J. Sebek traces the two "Fault15" lines to PLC-5 outputs 032 and 033 [R34], which WD-340-330-25 shows ending at the output cross-connect [R41]. The program makes them the main-focus and bucking-coil summaries, not repeats of the supplies' contacts: input 025 enters the main-focus summary and input 026 is not used [R38]. Table 29 matches each line to its PLC-5 point by the signal names of the panel schematic and of the digital and analog I/O drawings [R41], which agree; the drawings disagree only on pin and terminal numbers, which are checked by continuity (O2).

Table 29: Panel lines and PLC-5 points [R38], [R41].

Line (pin, SD-340-311-01)	Direction	PLC-5 point	Use in the program
HVPS Vout, Iout (J2-03, J2-04)	To PLC-5	Analog input module 2, channels 11 and 12 (terminals 13 and 14)	Collector power; HV detected above 400 counts
Main focus on (J2-06)	To panel	O:031/1	Filament requested and no main-focus overvoltage latch
Main focus Fault15 (J2-07)	To panel	O:031/12 (032)	Main-focus summary
Filament (J2-09)	To panel	O:031/3	Filament requested and no high-trip latch; the heater's FILAMENT_ON

Line (pin, SD-340-311-01)	Direction	PLC-5 point	Use in the program
Timeout (J2-10)	To panel	O:030/5	Warm-up complete or bypassed
HV detected (J2-11)	To panel	O:031/7	HVPS voltage above threshold
Local mode (J2-12)	To PLC-5	I:030/1	Read by EPICS only
Solenoid request (J2-18)	To PLC-5	I:031/3	With I:031/7, the online bit
Bucking coil on (J2-19)	To panel	O:031/2	Filament requested and no bucking-coil upper-limit latch
Bucking coil Fault15 (J2-20)	To panel	O:031/13 (033)	Bucking-coil summary
Filament request (J2-21)	To PLC-5	I:031/1	Starts the warm-up and enables the focus outputs
Timeout request (J2-22)	To PLC-5	I:031/2	Timer bypass, held while the filament is on
Reset request (J2-23)	To PLC-5	I:030/0	Clears the fault latches
RF detected (J2-24)	To panel	O:031/10	RF power above threshold
Online mode (J2-25)	To PLC-5	I:031/7	With I:031/3, the online bit
Area summaries (J3)	To panel	O:030/0 to /4 and /6 to /16; O:031/6	Cavity, klystron, focus, waveguide, air pressure, circulator, flow, vacuum, temperature; collector protection
AB summary (J3-09)	To panel	O:031/0	Global interlock sum
AB heartbeat (J3-21)	To panel	O:031/5	50 ms heartbeat

Table 30: RF Local Display Panel connector J2.

Pins	SD-340-311-01 label	WD-340-330-02 label
J2-01, J2-14	Vin+, Vin− (from HVPS)	HVPS Vin+, Vin−
J2-02, J2-15	Iin+, Iin− (from HVPS)	HVPS Iin+, Iin−
J2-03, J2-04	Vout, Iout (to AB)	HVPS Vout, Iout
J2-06	SolenoidM Ctrl24 (AB)	Main Focus On
J2-07	SolenoidM Fault15 (supply)	Main Focus PS Interlock
J2-09	Filament Ctrl24 (AB)	Filament On
J2-10	Timeout Ctrl24 (AB)	Filament Timer Bypass
J2-11	HV Detected24 (AB)	HV Detected
J2-12	Local Mode24 (AB)	Local/Remote
J2-18	Solenoid Req24	Online/Offline Request
J2-19	SolenoidB Ctrl24	Bucking Coil PS On/Off
J2-20	SolenoidB Fault15 (supply)	Bucking Coil PS Interlock
J2-21	Filament Req24	Filament PS On/Off
J2-22	Timeout Req24	Filament Timer Bypass
J2-23	Reset Req24	Interlock Reset
J2-24	RF detected24	RF On Indicator
J2-25	Online mode24	Online/Offline Control

Pins	SD-340-311-01 label	WD-340-330-02 label
J2-05, -08, -13, -16, -17	Returns	Returns

Connector J3 carries 24 V summary lines from the PLC-5 (SD-340-311-01 pins): cavity 1 to 4 (J3-01, J3-14, J3-02, J3-15), klystron (J3-03), focus (J3-04), magic tee 1 to 3 (J3-17, J3-05, J3-18), waveguide pressure (J3-06), water flow (J3-07), temperature (J3-08), collector (J3-16), circulator (J3-19) and vacuum (J3-20); the AB summary (J3-09) and the AB heartbeat (J3-21); spares on J3-10, J3-11 and J3-22 to J3-25; ground on J3-12 and J3-13. The cross-connect diagram numbers the same signals in reverse pin order (for example, its J3-05 “Interlock Sum” corresponds to J3-09 and its J3-18 “AB Heartbeat” to J3-21), and it assigns “Waveguide Air MPS” to the line that the schematic shows as spare 1 (J3-22); J. Sebek records the same discrepancy [R34]. Any connection at J3 is checked on the hardware first.

D Abbreviations

Abbreviation	Meaning
AB	Allen-Bradley
ADSO	Accelerator Directorate Safety Office (SLAC)
AIM	Arc Interlock Module (VXI slot 12)
CLK	VXI clock and RF reference distribution module
DCM	Direct Communication Module (1771-DCM, 1747-DCM)
EPICS	Experimental Physics and Industrial Control System
Fast IC	Fast Interlock Chassis 340-308
GAL	Generic array logic, a programmable logic device
HV, HVPS	High voltage, high-voltage power supply
IOC	EPICS Input/Output Controller
IQA	In-phase, quadrature and amplitude detector module
LLRF	Low-level RF
MPS	Machine protection system
PPS	Personnel protection system
PV	EPICS process variable
RFP	RF Processor module
RIO	Allen-Bradley Remote I/O
SNL	State Notation Language (EPICS sequencer)
SSA	Solid-state amplifier
TID	Technology and Innovation Directorate
VXI	VME eXtensions for Instrumentation